

The hadron energy resolution of a dual-readout calorimeter using a Kaon/Proton beam in the 2024 test beam experiment

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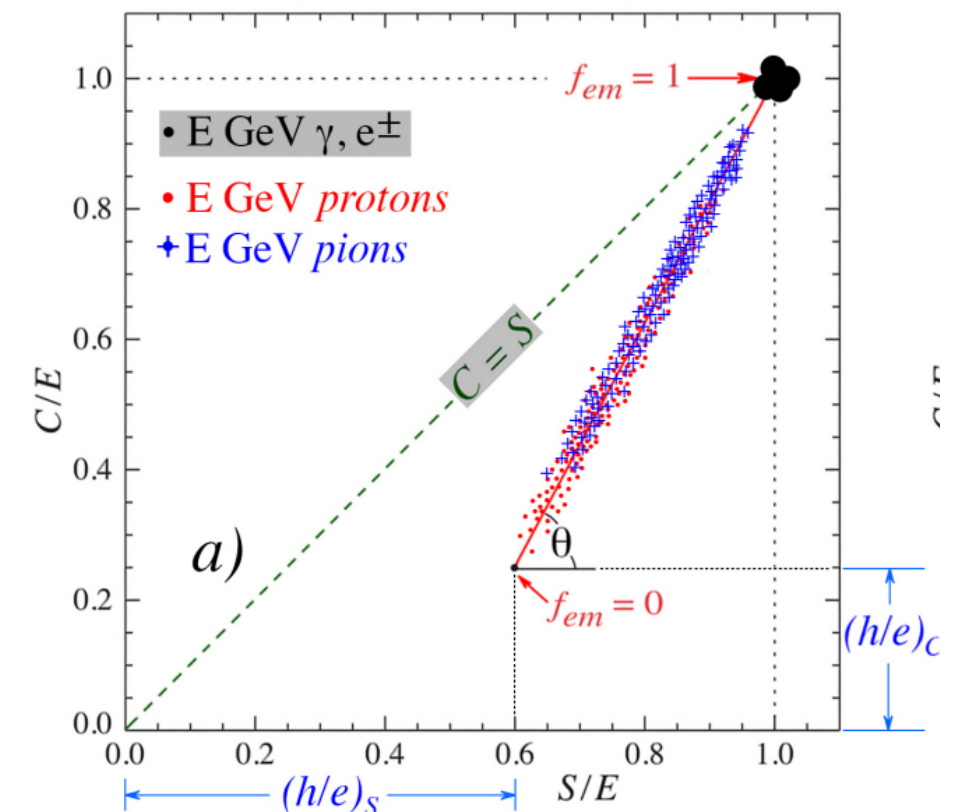
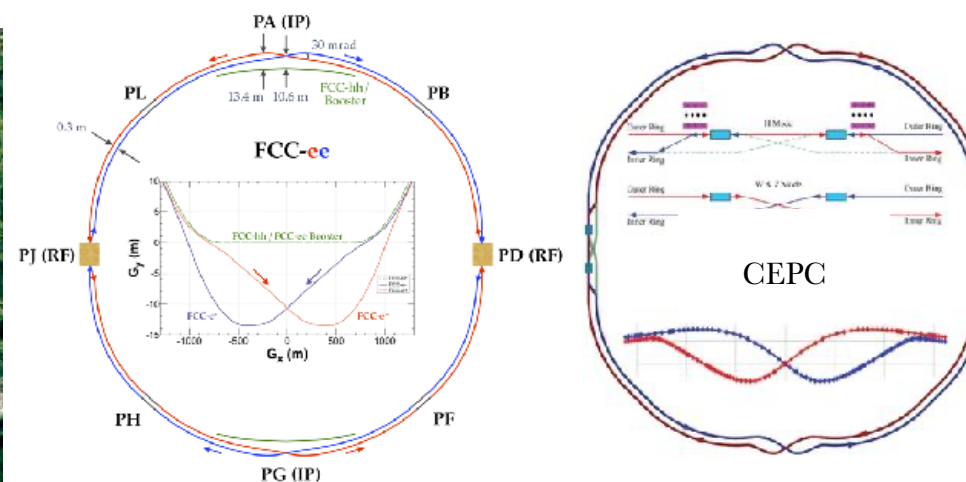
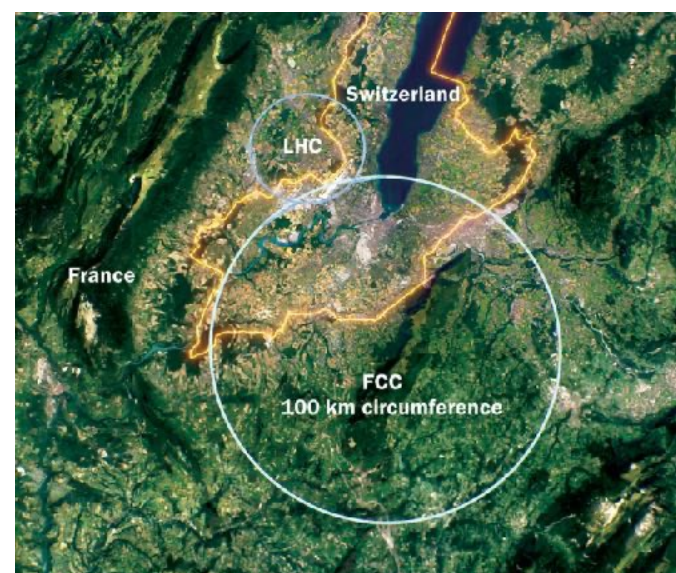
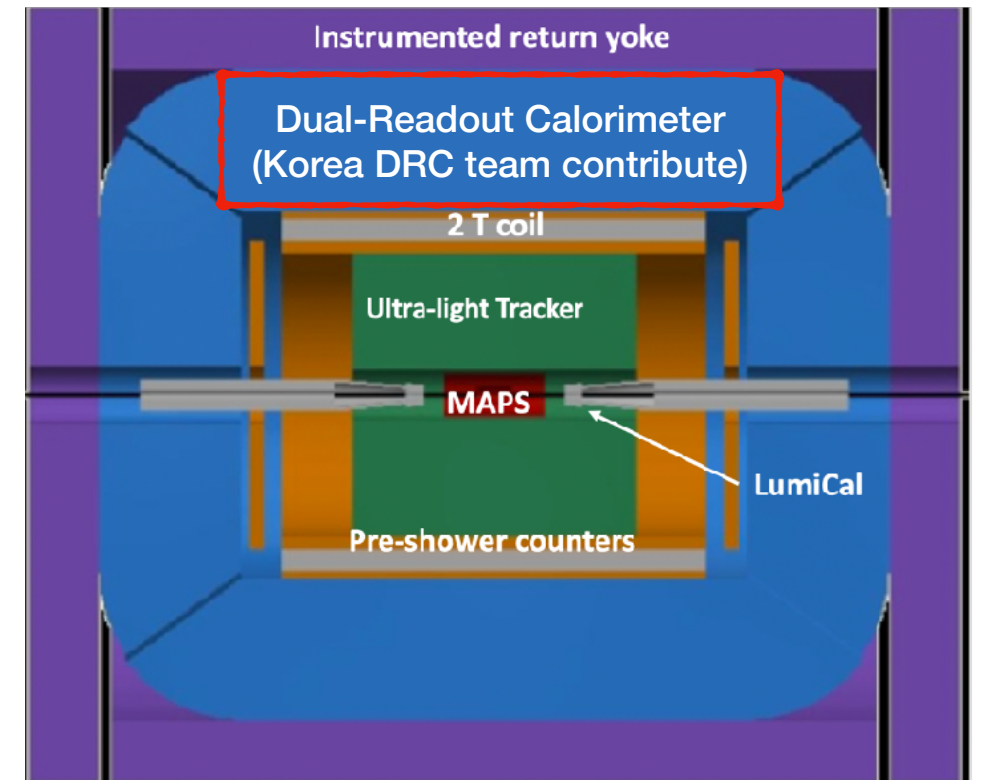
On behalf of the Korea Dual-Readout Calorimeter Team



Supported by



- **Dual-Readout Calorimeter**
 - The dual-readout calorimeter has been included in the conceptual design report of both FCC-ee and CEPC
 - **Non-gaussian electromagnetic fluctuations** are a major factor that makes it difficult to measure the energy of a hadron shower
 - The dual-readout calorimeter offers **high-quality energy measurement** for both EM particles and hadrons simultaneously
 - Outstanding energy resolution can be achieved by **measuring EM components** and **correcting hadron energy event by event**



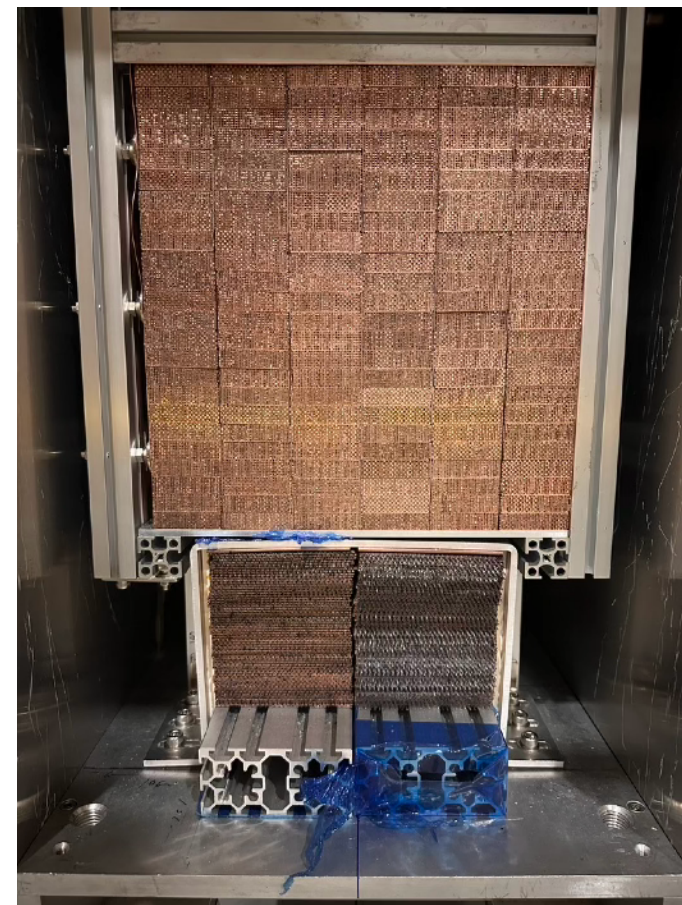
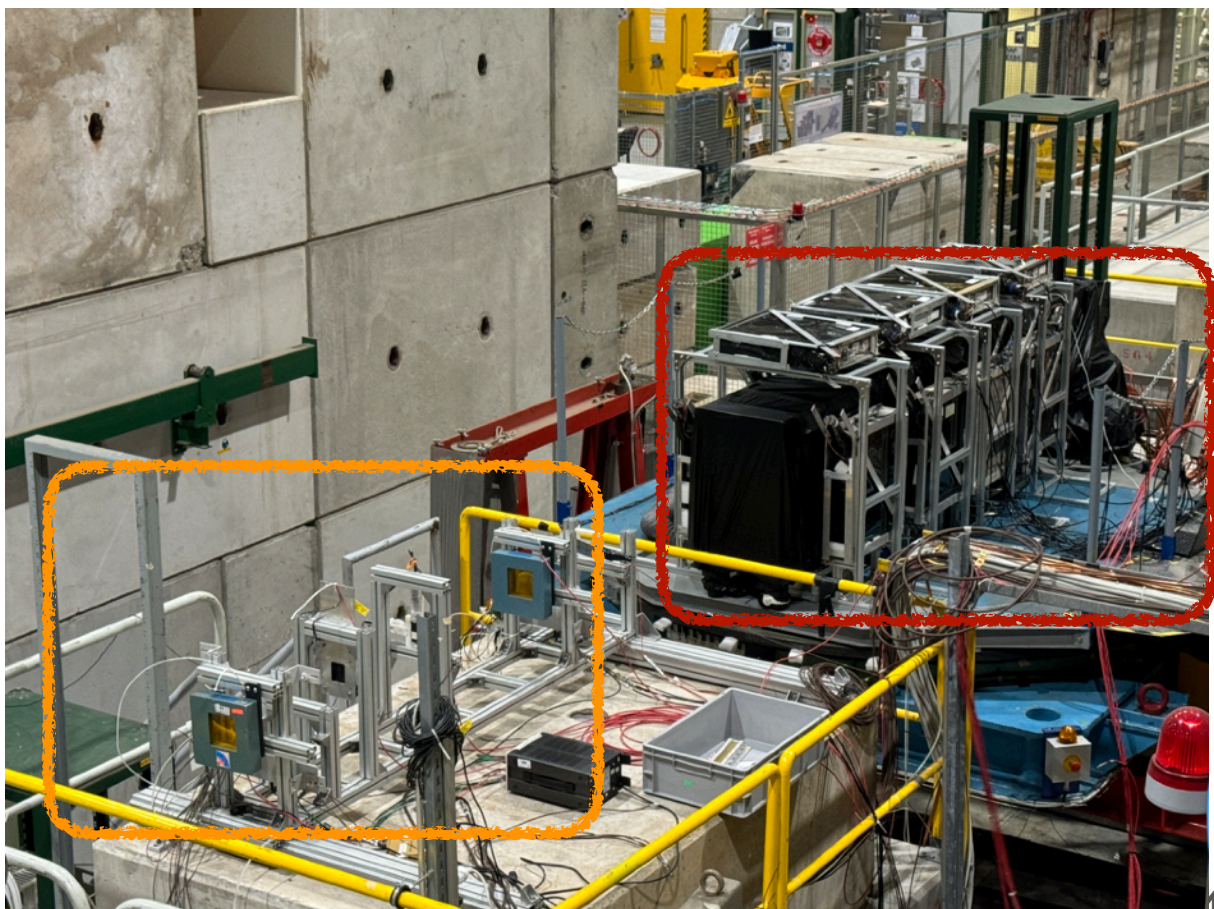
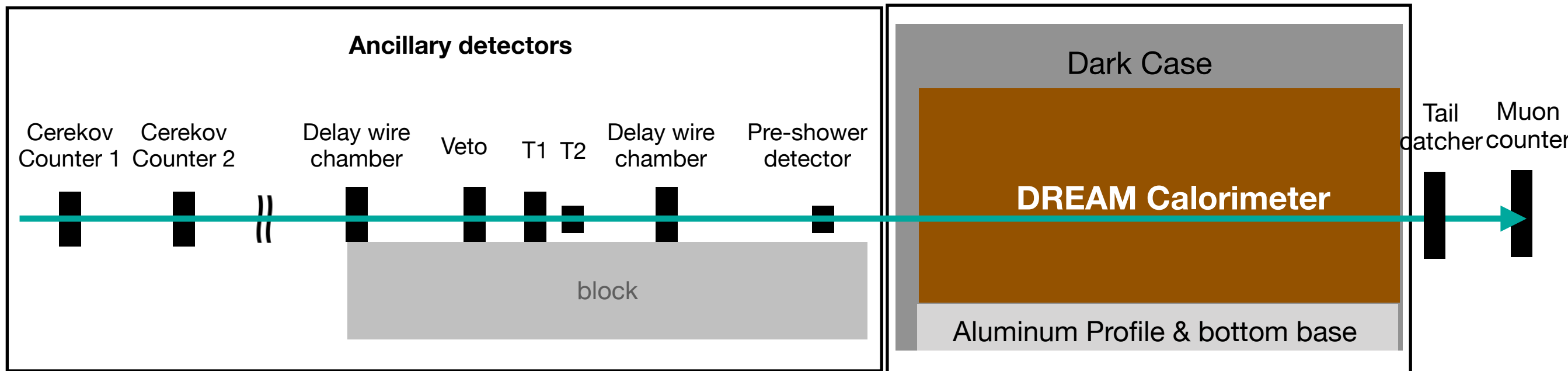
Response of Calorimeters to Different Hadrons

- π :
High electromagnetic component due to frequent π^0 production, resulting in a strong Cherenkov signal
Remaining energy in the pion allows it to act as a **leading particle**, continuing to interact and lose energy
- p :
 - Limited π^0 production due to **baryon number conservation**, leading to low electromagnetic component and a weak Cherenkov signal
 - Protons retain **some energy** after collision, allowing them to become **leading particles** and continue interacting in the calorimeter
- K :
 - Kaons, being heavier mesons that contain strange quarks, can undergo both **strong and weak interactions**, leading to more complex decay processes. Kaons produce a moderate Cherenkov signal and exhibit combined hadronic and electromagnetic showers
 - **Kaons with remaining energy** can act as **leading particles**, continuing interactions in the calorimeter

Test Beam Setup

- In the test beam experiment, we used auxiliary detectors such as DWC, trigger, veto trigger, and pre-shower detector

Details are presented @ Seoyun Jang's talk(J2.04) & Geunpil An talk(J2.08)



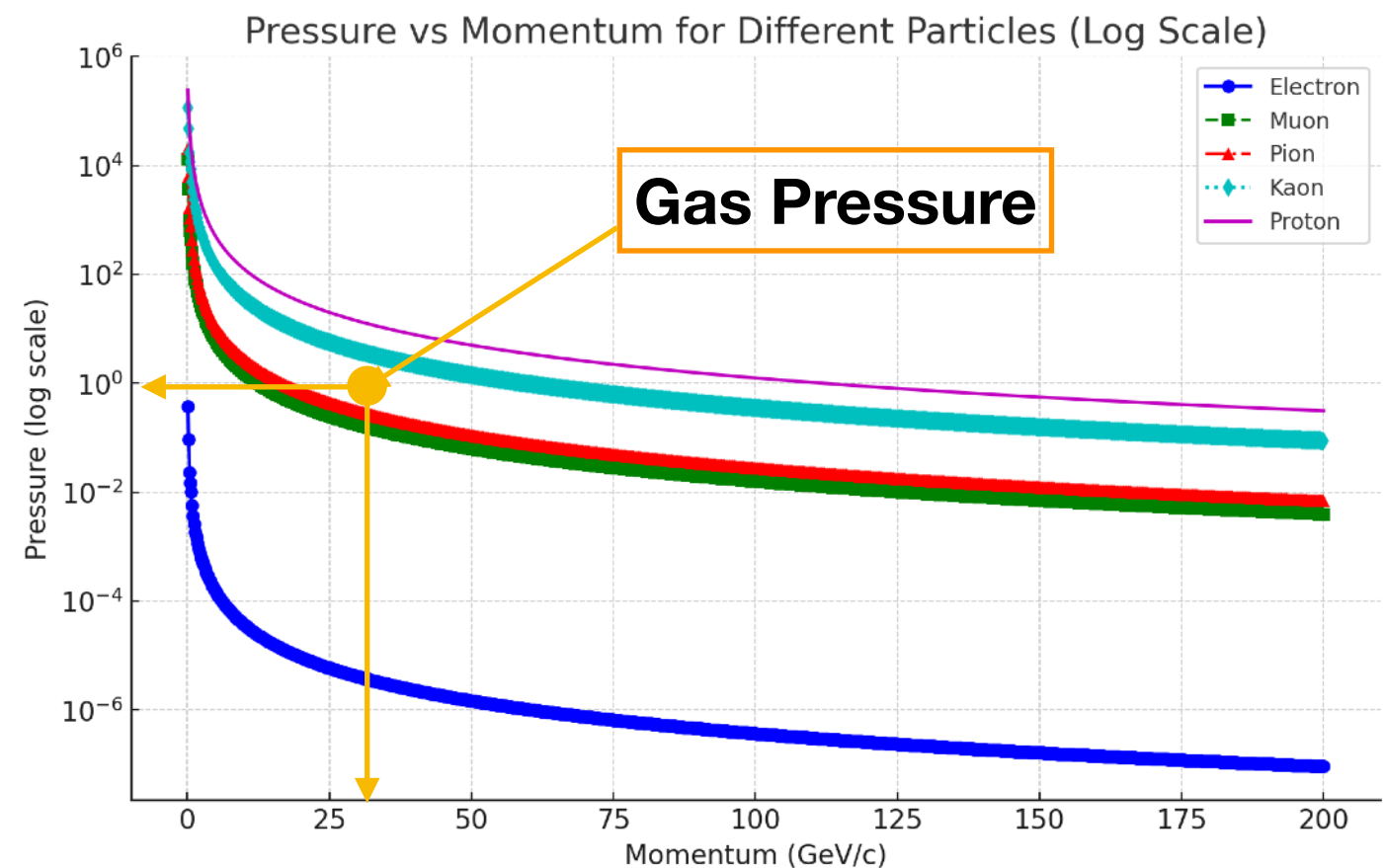
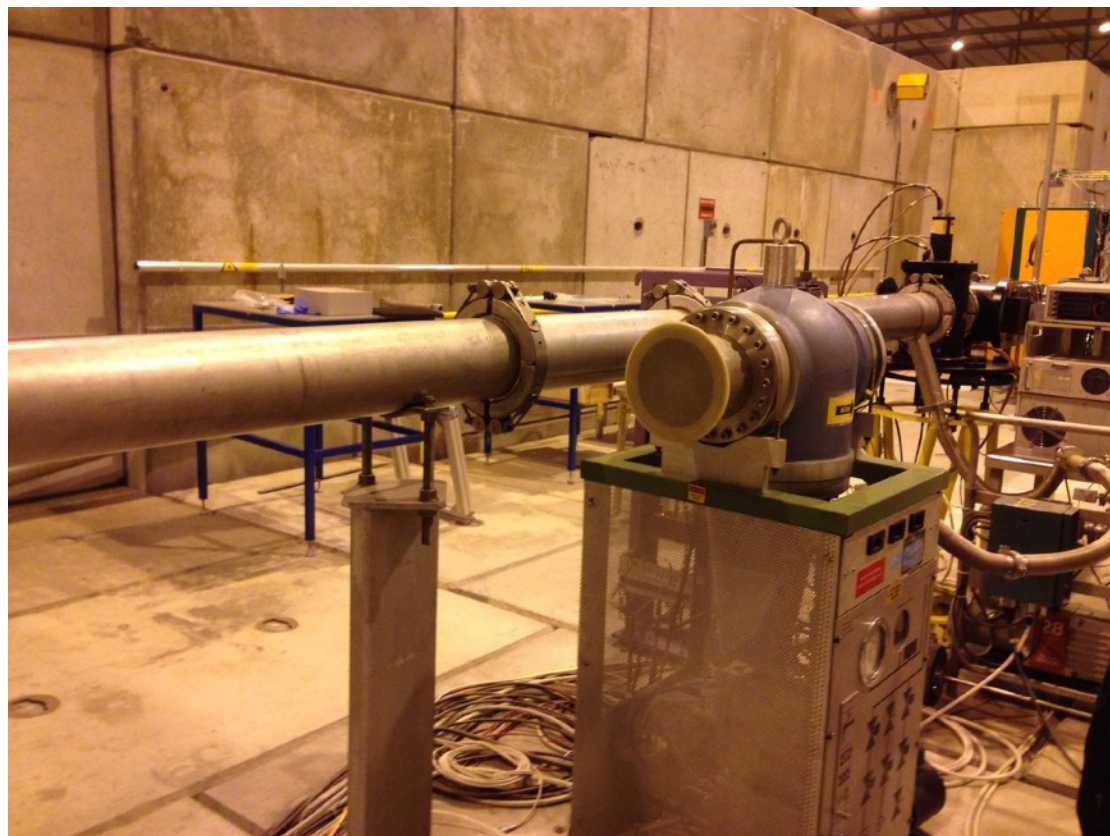
Run List

- For the DRC study on energy resolution using **Kaons and Protons**, we used the runs listed in the table below (mixed hadron beam)

Beam Type	Target Tower	Energy (GeV)	Run	# of event (wave)
K/p	M5T1	20	10650	14059
		40	10649	70000
		60	10648	100000
		80	10647	100000
		100	10646	100000
		120	10645	50000
Pion	M5T1	20	10643	16000
		40	10641	70000
		60	10638	150000
		80	10635	100000
		100	10631	50000
		120	10616	50000

How to Separate Hadron Beams (i)

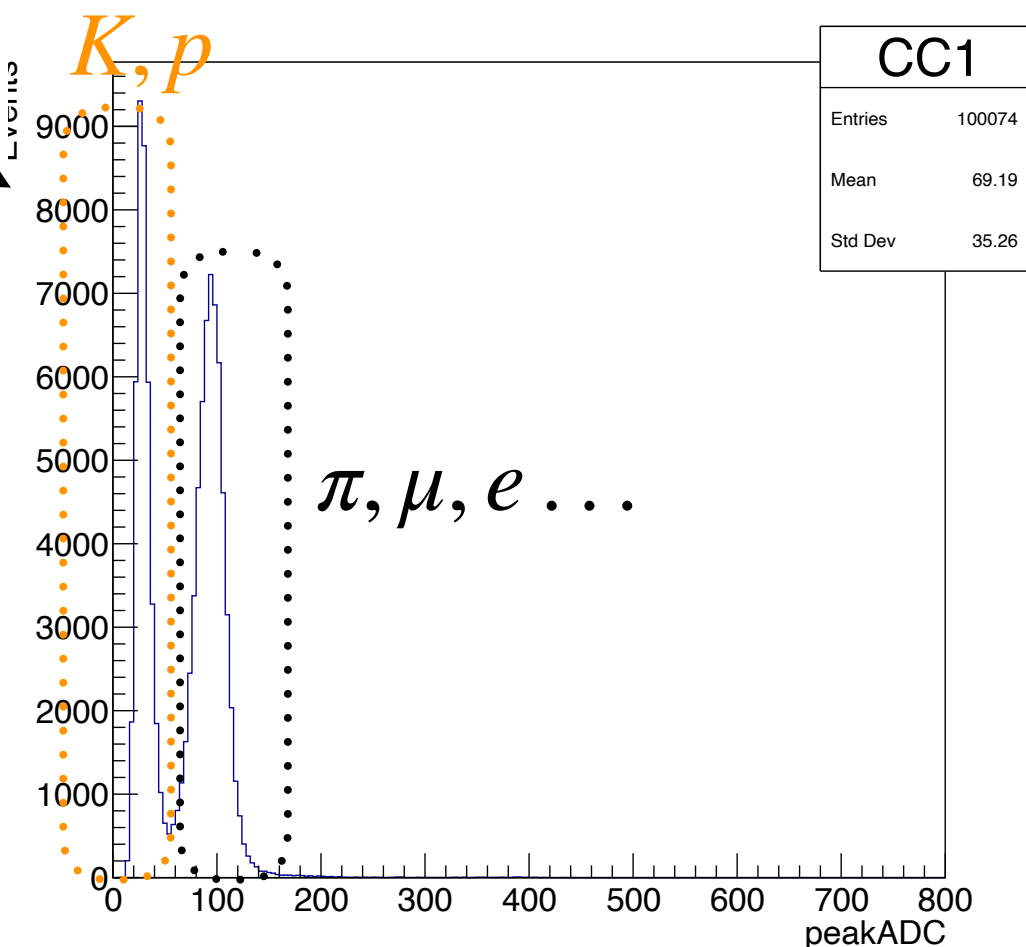
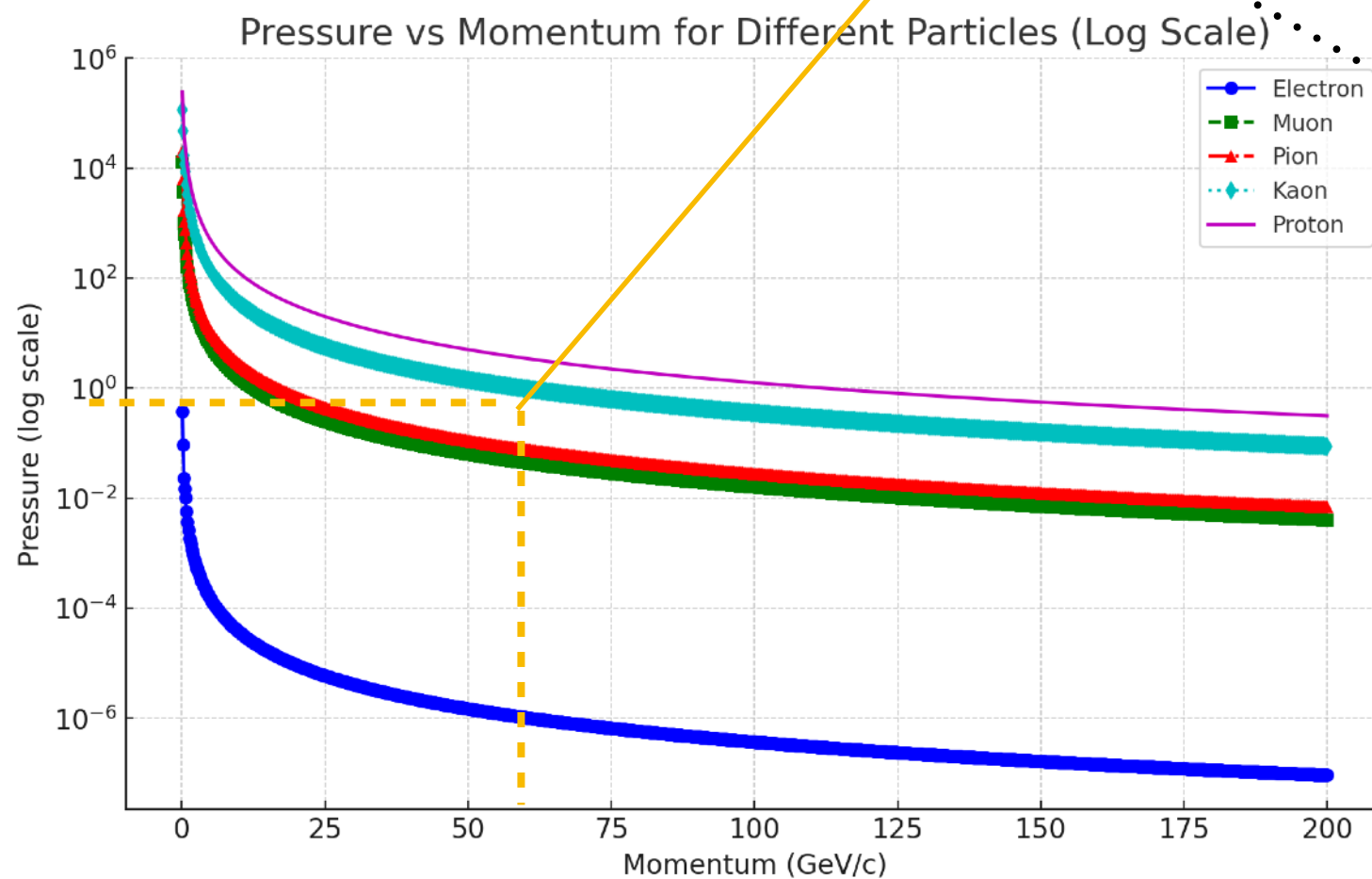
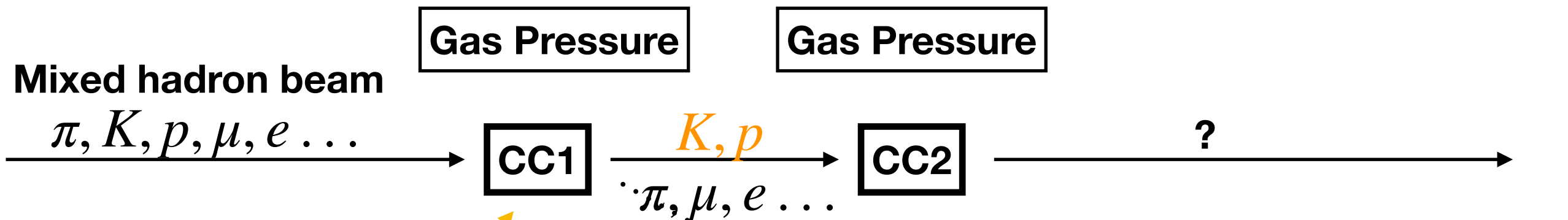
- Since SPS provided mixed hadrons, we have to separate hadron particles
- Cherenkov counters detect **charged particles** moving **faster than the speed of light** in a medium by measuring the emitted **Cherenkov radiation**
- The **gas pressure** in the counter controls the detection threshold, allowing for particle identification based on velocity
- This setup helps **distinguish** between particles like protons, pions, and kaons at different momenta



How to Separate Hadron Beams (ii)

- We used two Cherenkov counters, and by using different pressures in each counter, we can distinguish between kaons and protons

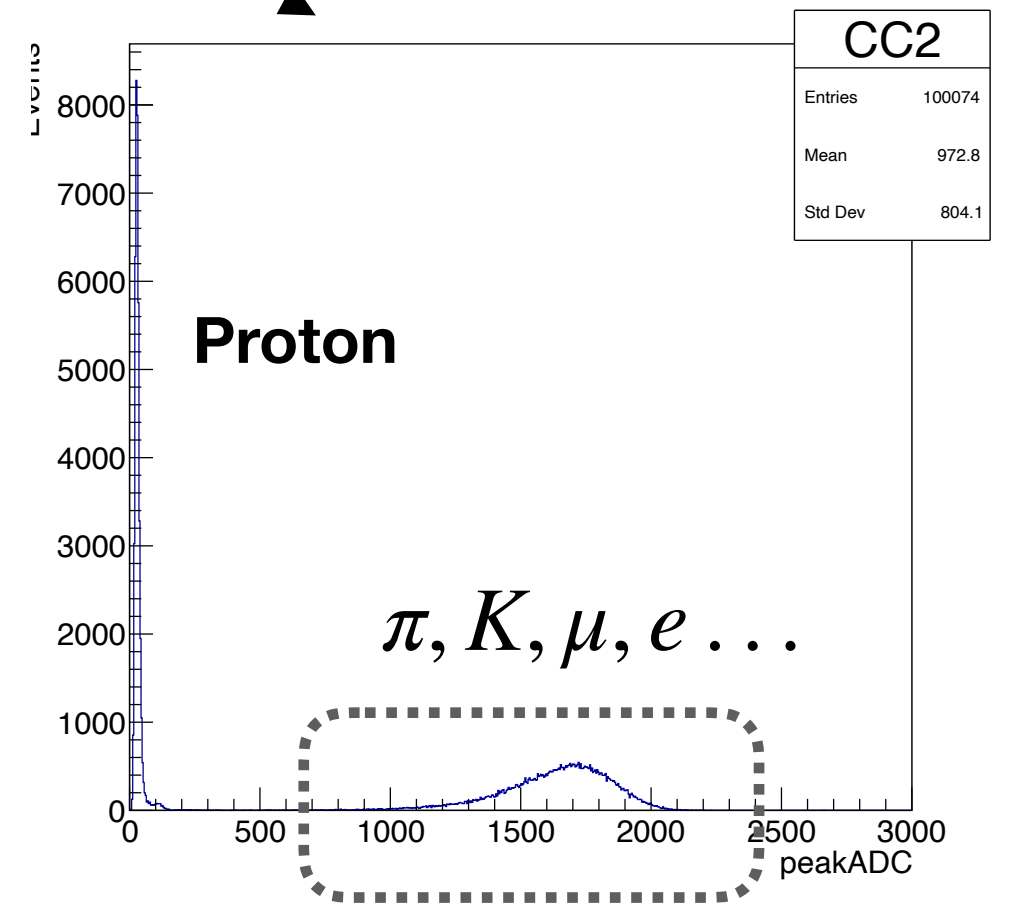
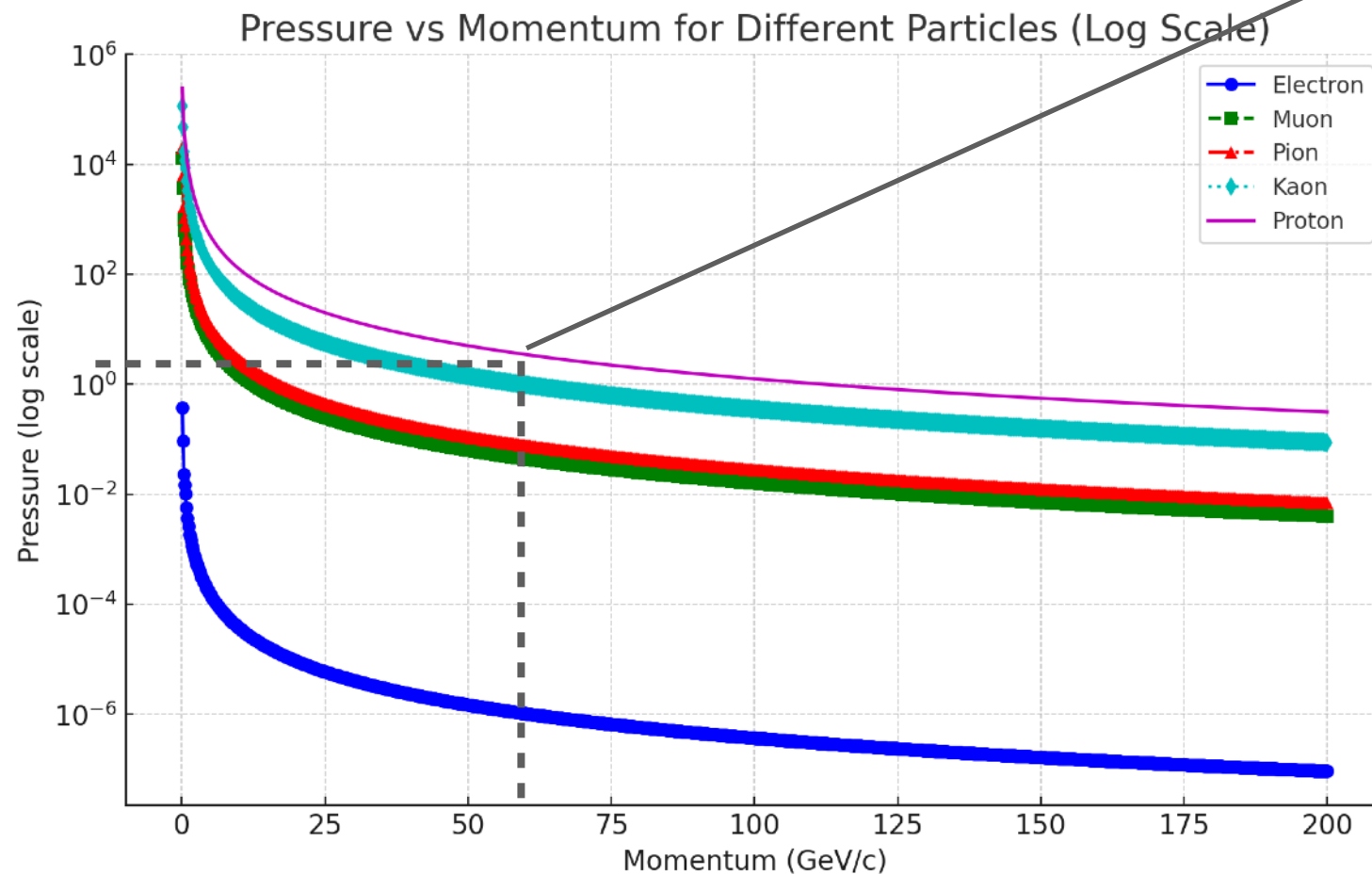
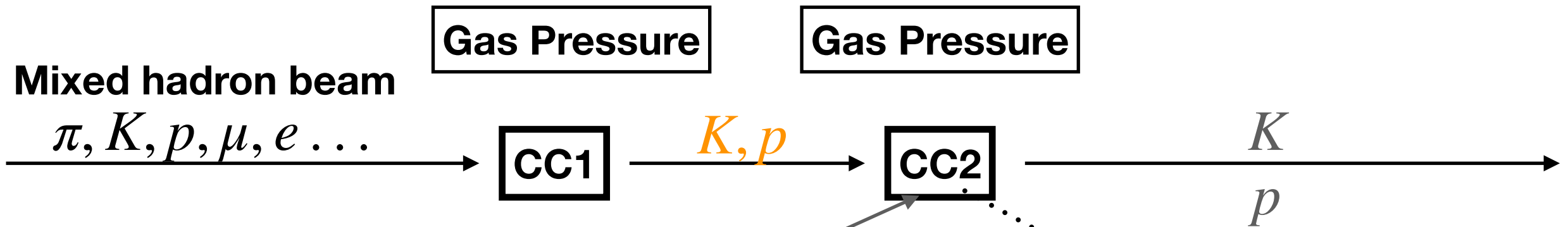
Ex) Kaon selection



How to Separate Hadron Beams (iii)

- We used two Cherenkov counters, and by using different pressures in each counter, we can distinguish between kaons and protons

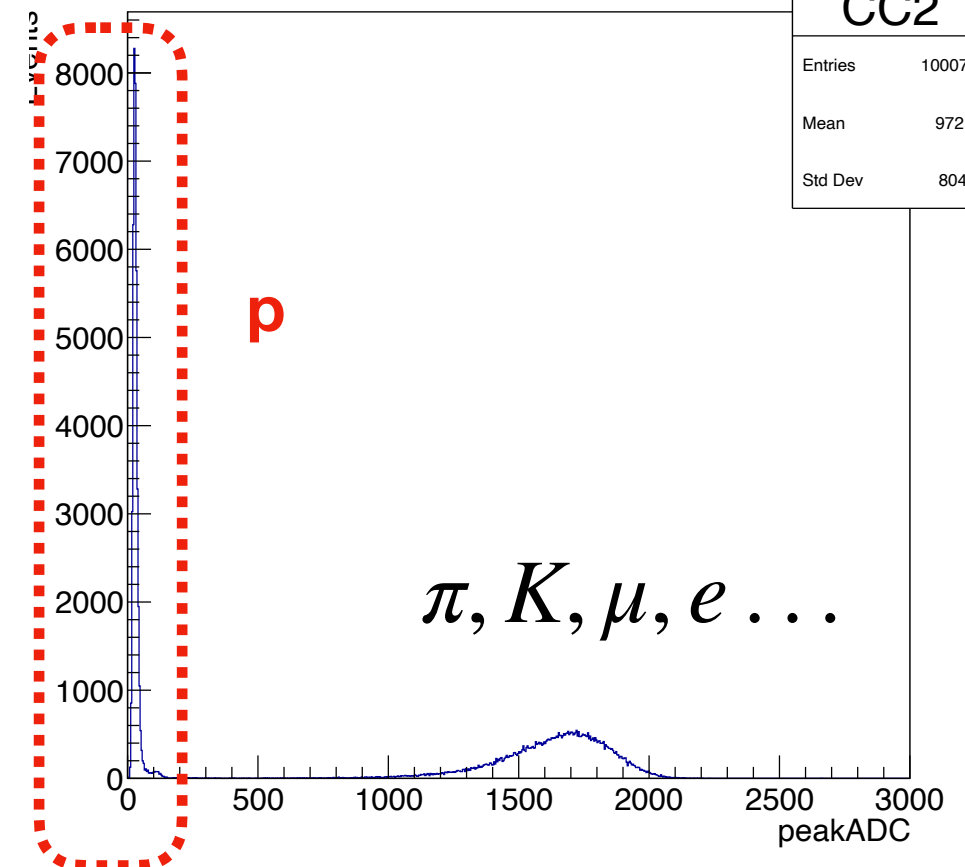
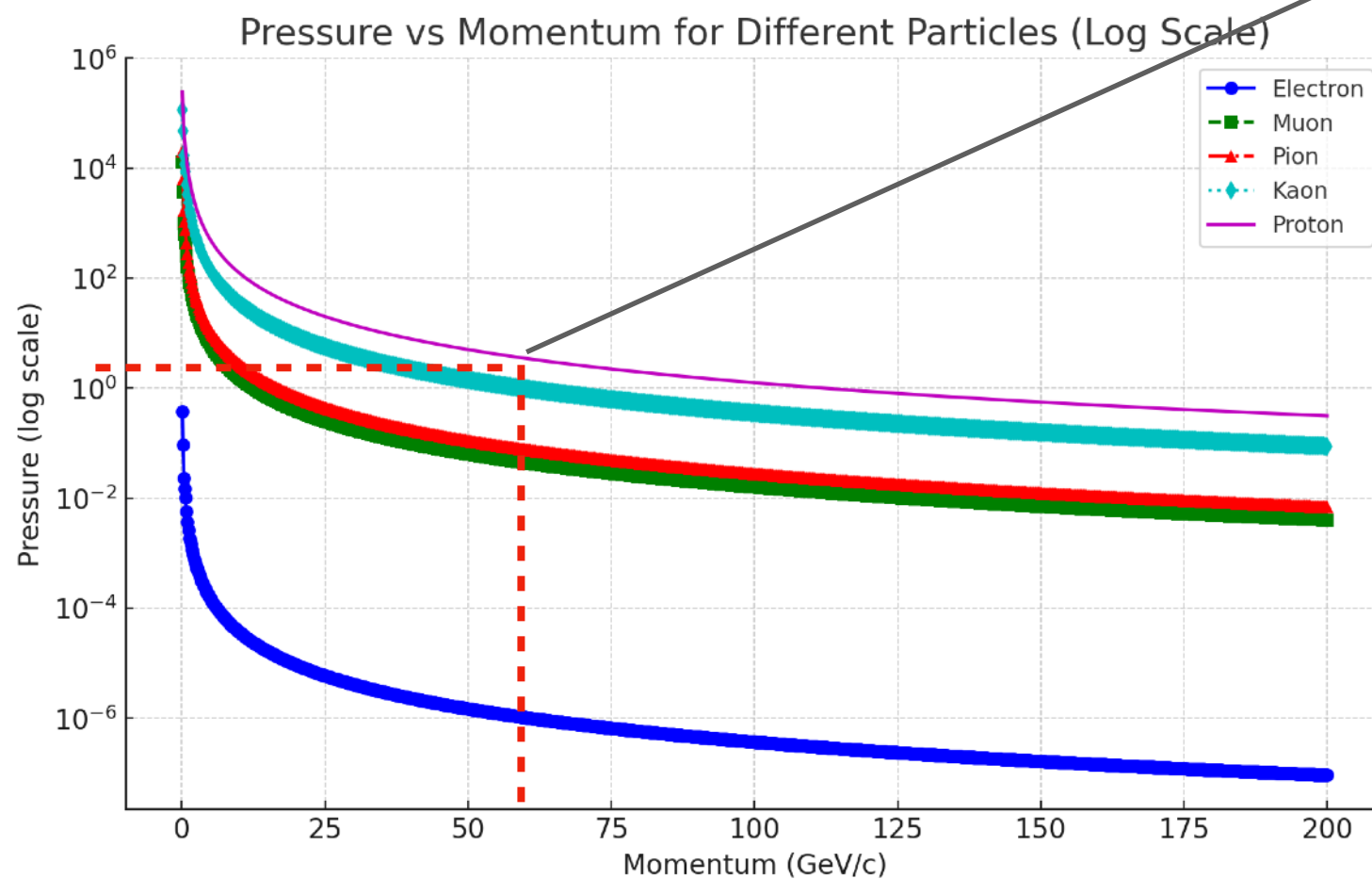
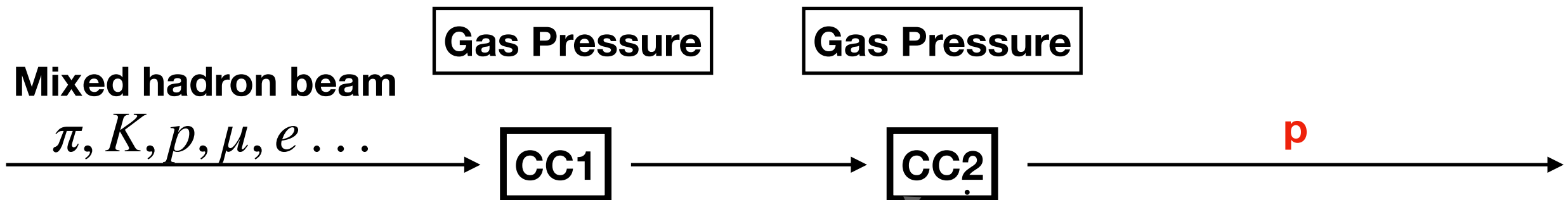
Ex) Kaon selection



How to Separate Hadron Beams (iV)

- We used two Cherenkov counters, and by using different pressures in each counter, we can distinguish between kaons and protons

Ex) p selection



Events Selection

- The event selection was primarily and need to updated
 - using CC & Ancillary detectors, the beam type(Pion/Kaon/Proton) are separated

	Pion	Kaon	Proton
DWC	$ \text{DWC1} - \text{DWC2} < 2\text{mm}$ 1 x 1 cm beam spot		
CC1	-	$50 > \text{Peak}$	-
CC2	$150 > \text{peakADC}$	$150 > \text{peakADC}$	$150 < \text{peakADC}$
PS (Remove electron)	$80 < \text{peakADC} < 300$ (Select 1 mip peak)		
MC (Remove muon)	$\text{peakADC} < 30$ (Reject moon candidate signals)		

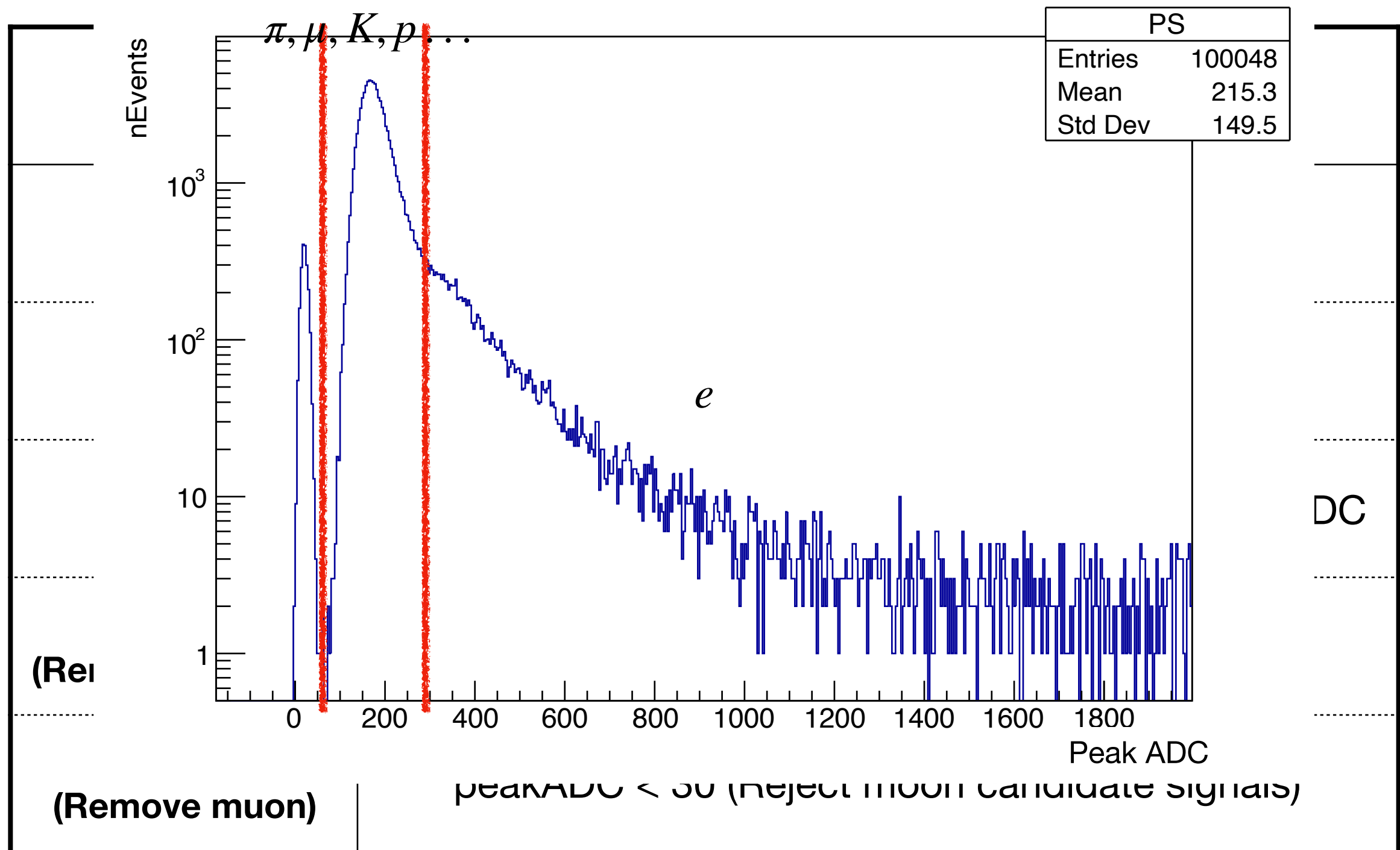
Events Selection

- The event selection was primarily and need to updated

- usi

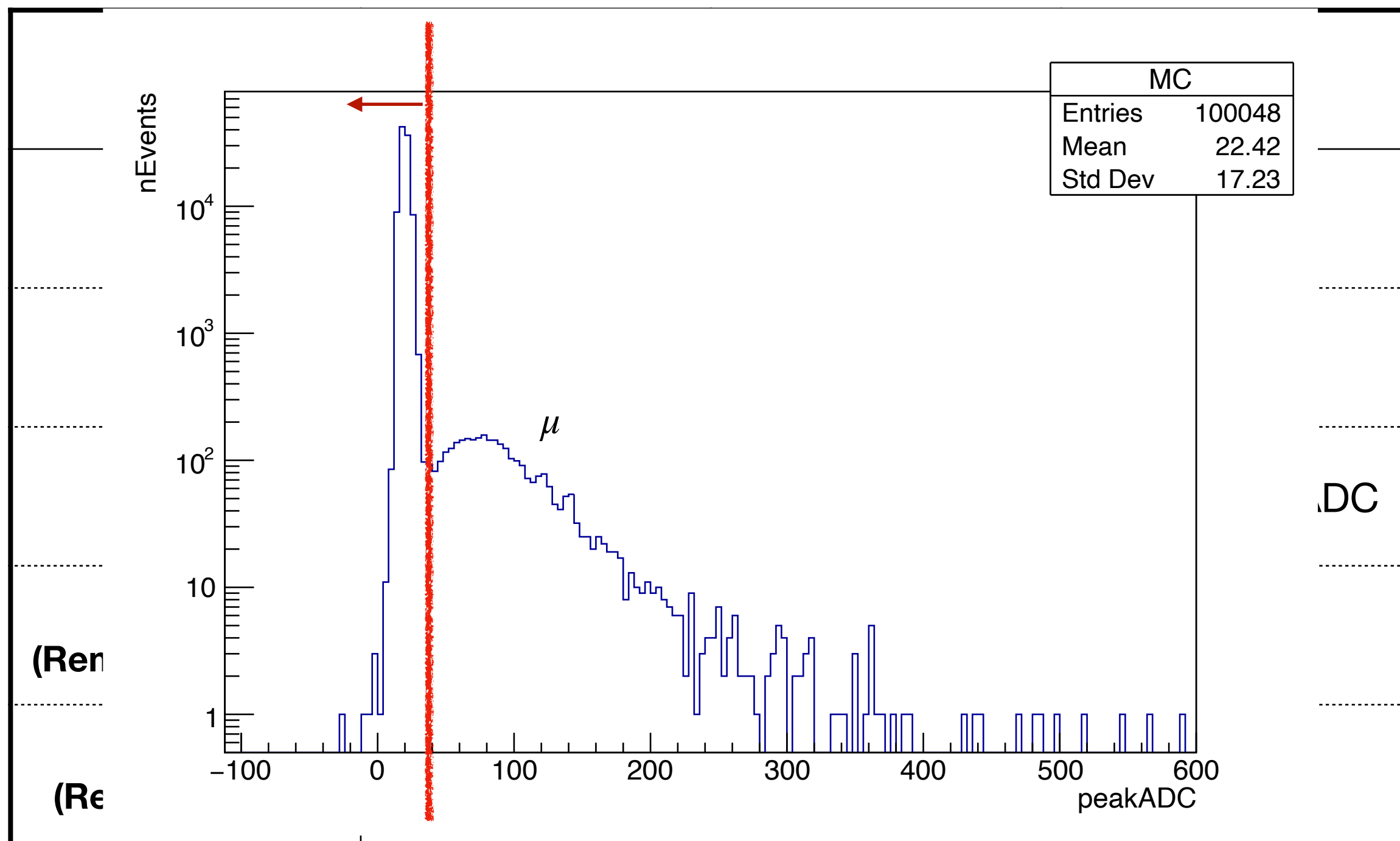
Select

arated



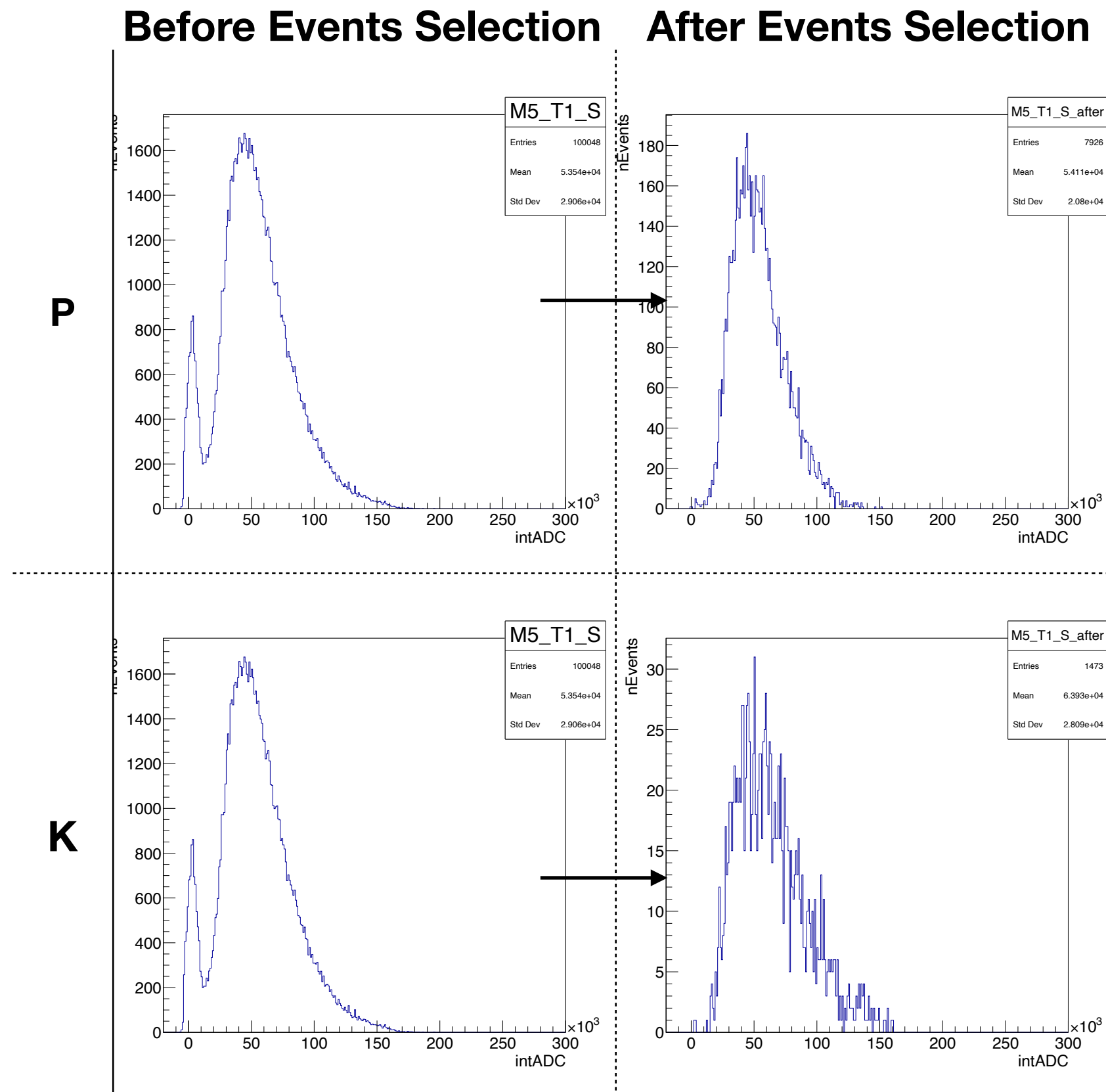
Events Selection

- The event selection was primarily and need to updated
 - using CC & Ancillary detectors, the beam type(Pion/Kaon/Proton) are separated



Int. ADC distributions for M5T1

- Using auxiliary detectors, we selected protons (kaons)
- For protons(kaons) at 80 GeV, approximately 8k(1.5k) events out of 100k are proton candidate events, resulting in a selection efficiency of about 8%(1.5%)



Calibration & DR correction

- Calibration

- The purpose of equalization is to ensure that all calorimeter towers produce consistent signals for the same energy deposition (with EM particles; positron)

Details are presented @ Kyeongpil Lee's talk(J2.02)

- Dual-readout correction

- One of the key advantages of a DRC is its ability to correct for fluctuations in the electromagnetic and hadronic components of a shower by simultaneously reading out the Cherenkov and scintillation signals

$$E = \frac{S - \chi C}{1 - \chi}$$

Details are presented @ Sungwoon Kim's talk(J2.03)

- Energy Leakage in DRC

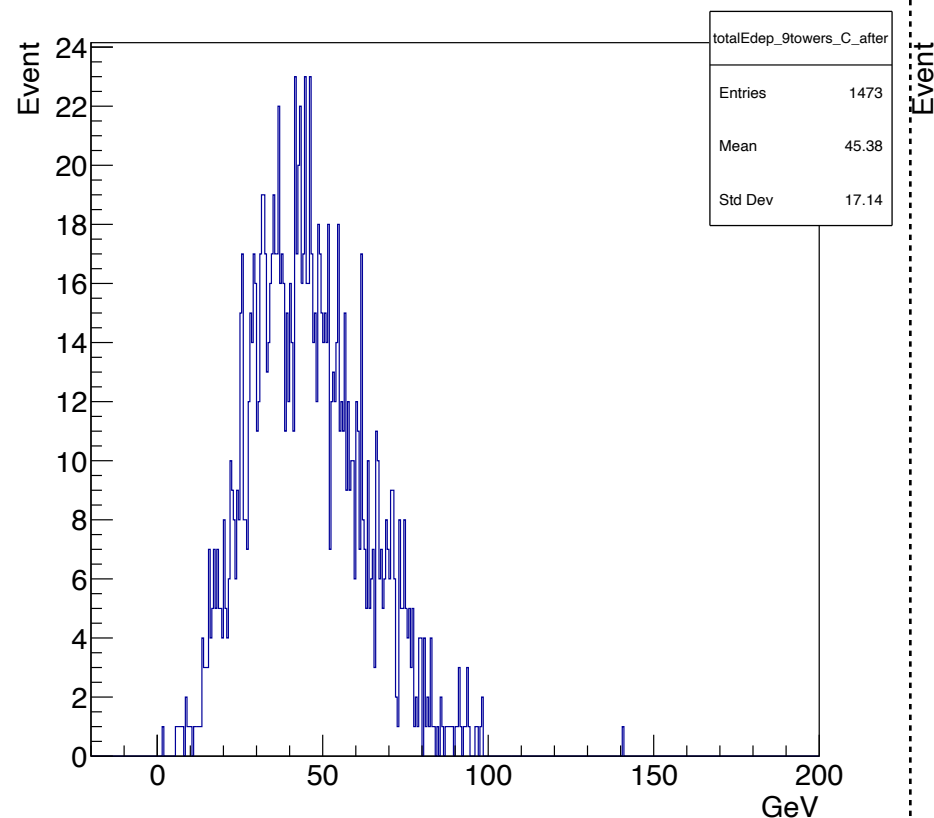
- **10% energy leakage** for hadronic shower
- To measure this leakage, we installed **leakage counters** around the DRC
- The leakage counters are equipped with large **scintillators** that allow us to measure the energy of the leaking particles

Details are presented @ Sungwoon Kim's talk(J2.03)

Energy Distribution (C, 80GeV)

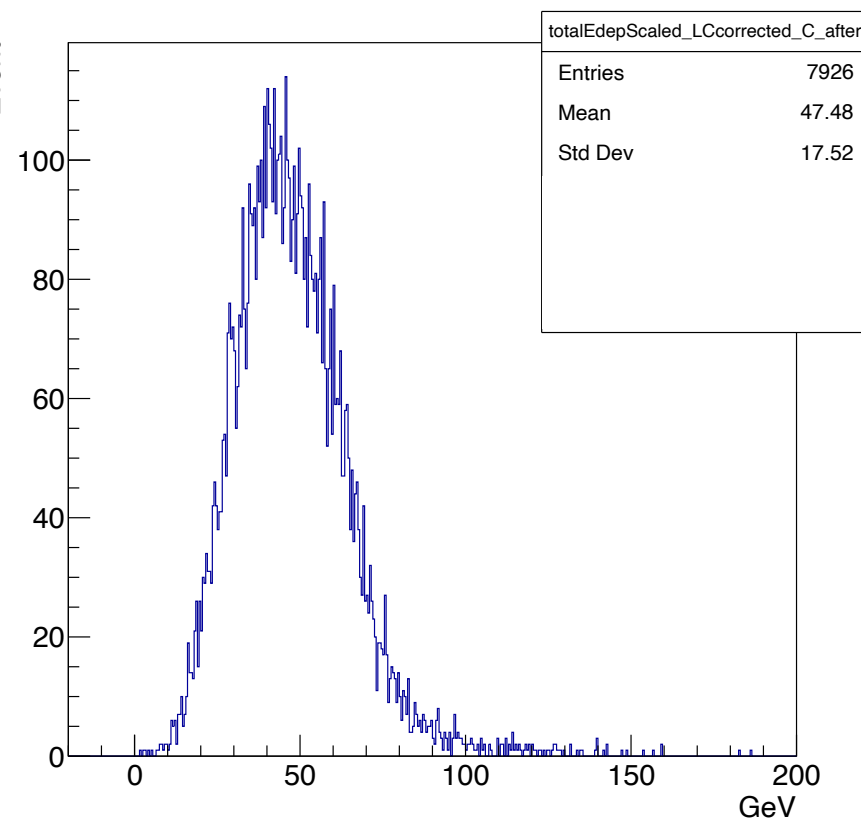
Very preliminary results !!!!

Kaon



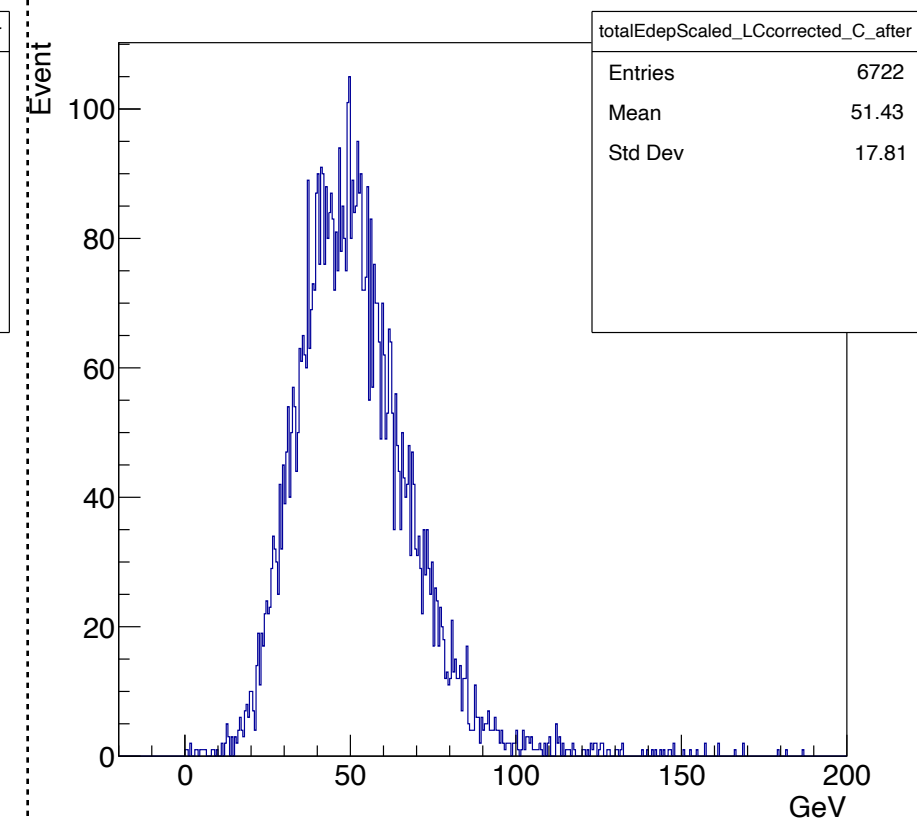
$rms/mean = 33.8 \%$

Proton



$rms/mean = 36.9 \%$

Pion

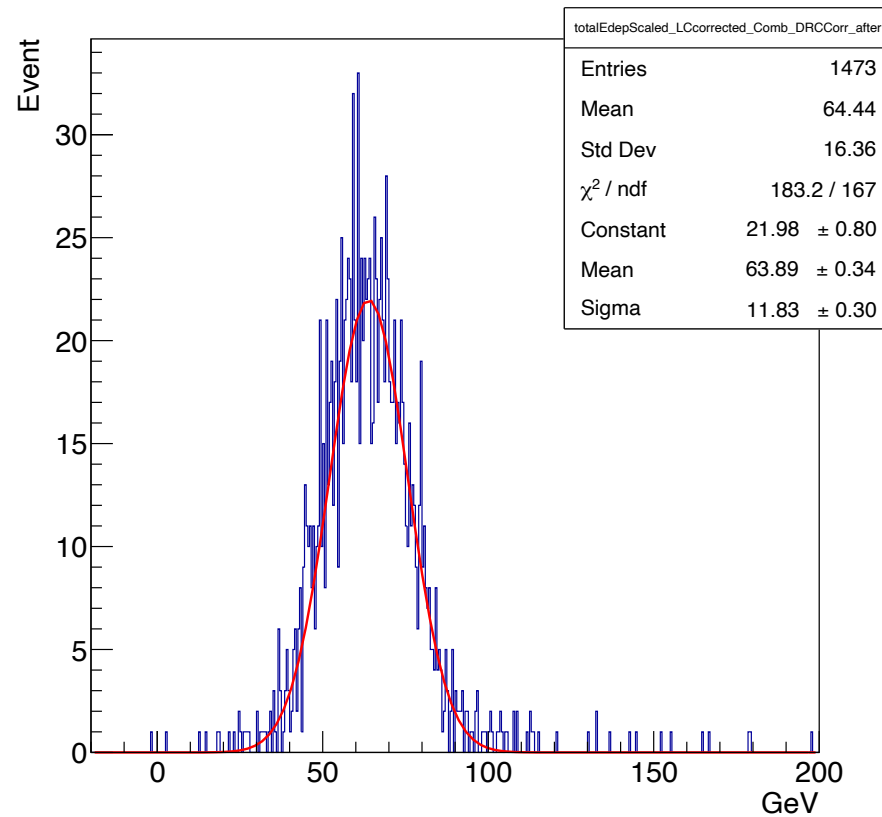


$rms/mean = 34.6 \%$

- Differences in response from hadrons arise from the structure and interaction behavior of the particles, where more electromagnetic content corresponds to a stronger Cherenkov signal (8% difference between p & π)

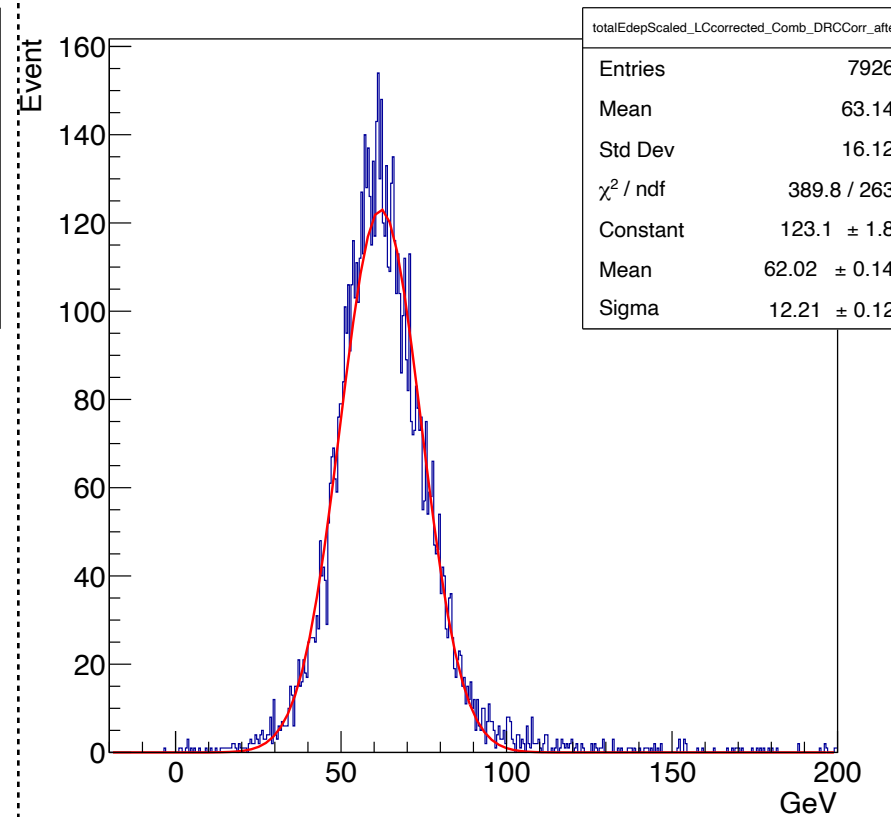
Very preliminary results !!!!

Kaon



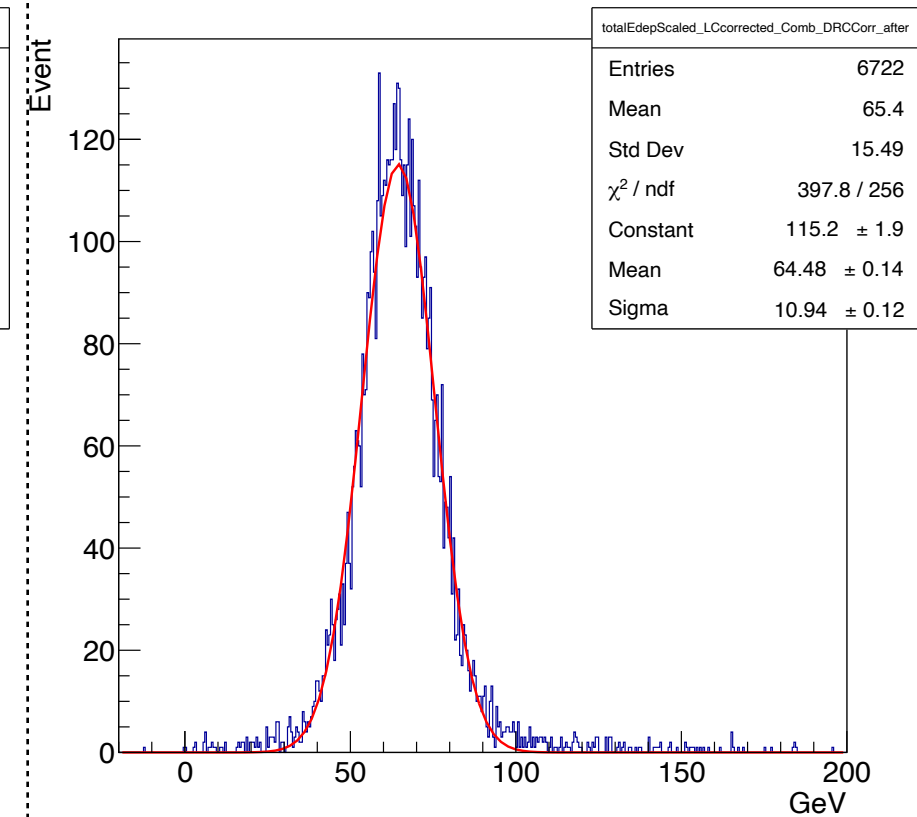
$$\sigma/E = 18.52\%$$

Proton



$$\sigma/E = 19.7\%$$

Pion



$$\sigma/E = 17.0\%$$

- After applying the **dual-readout correction**, which combines the cerenkov and scintillation signals, the differences between protons and pions are largely eliminated

The dual-readout method corrects for the differences in the electromagnetic and hadronic components of the showers, leading to similar signal distributions for both particles

Summary

- We conducted an experiment using hadrons in the test beam this year
- Cherenkov counters were used to distinguish between kaons and protons
- This marks the first hadron experiment performed with the full-size DRC(copper + fiber based) from Korea Dual-Readout Calorimeter Team
- We will derive the resolution results for the remaining energy points (20GeV ~ 120GeV; except for 80GeV)
- Additionally, we plan to optimize further the requirements related to the PID values or selection criteria



Backup



2024 Test Beam: SPS

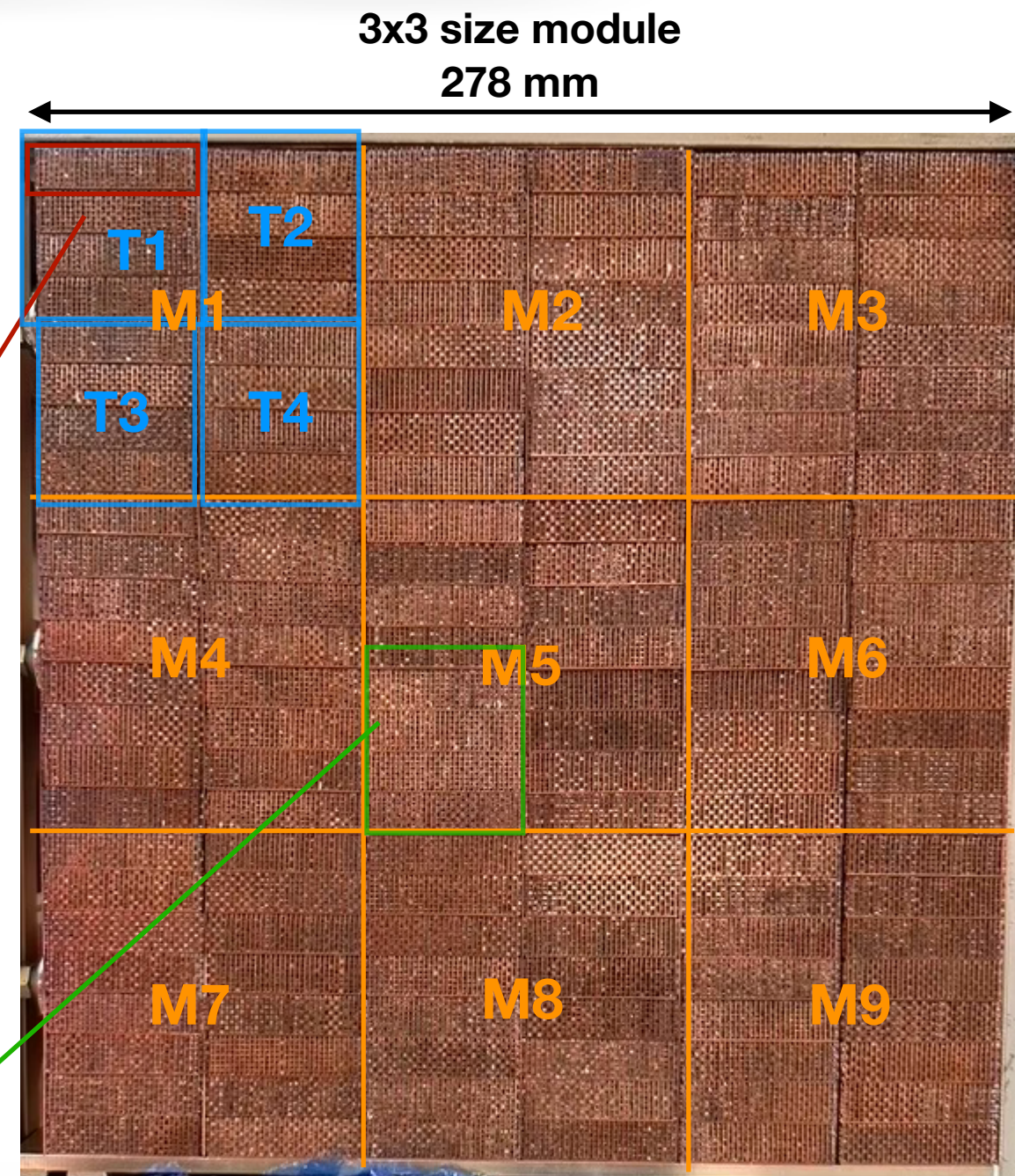
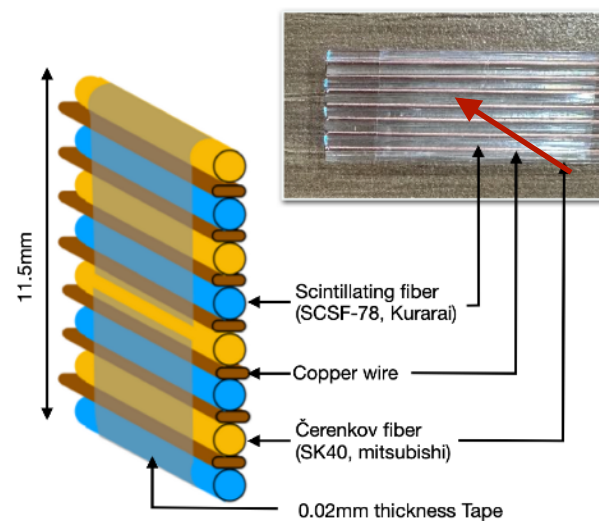
- **Duration : Aug. 6th ~ 28th**
- **Beam:**
 - e^+ beam : 6, 10, 20, 30, 40, 60, 80, 100, and 125 GeV
 - π beam : 20, 60, 80, 180 GeV
- **Goal:**
 - Verified characteristics of the full-size module
 - Measured EM & hadron energy performance for each produced module
 - Evaluated performance under various configurations for each module and tower with 2022 modules
 - Toward TDR!
- **Experimental hall**
 - During test beam, our experiments conducted at T4-H8 @ North Area



Module Configuration : TB2024

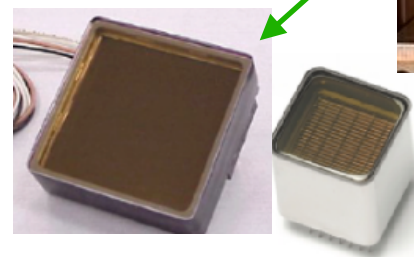
• Configuration

- Stack Configuration: 30 fiber layer units(4S+4C+7Cu wire) per stack
- Tower Configuration: 4 stacks per tower, totaling 120 fiber layer units
- Module Configuration: 4 towers per module, totaling 16 stacks or 480 fiber layer units
- 3x3 Prototype Configuration: 9 modules, totaling 4608 fiber layer units (including gap layers between copper blocks)



• Readout information

- 2 MCP-PMT
 - PLANACON XP85012, XP85112
- 35 Square shape PMT (Generic PMT)
 - Hamamatsu R11265-100

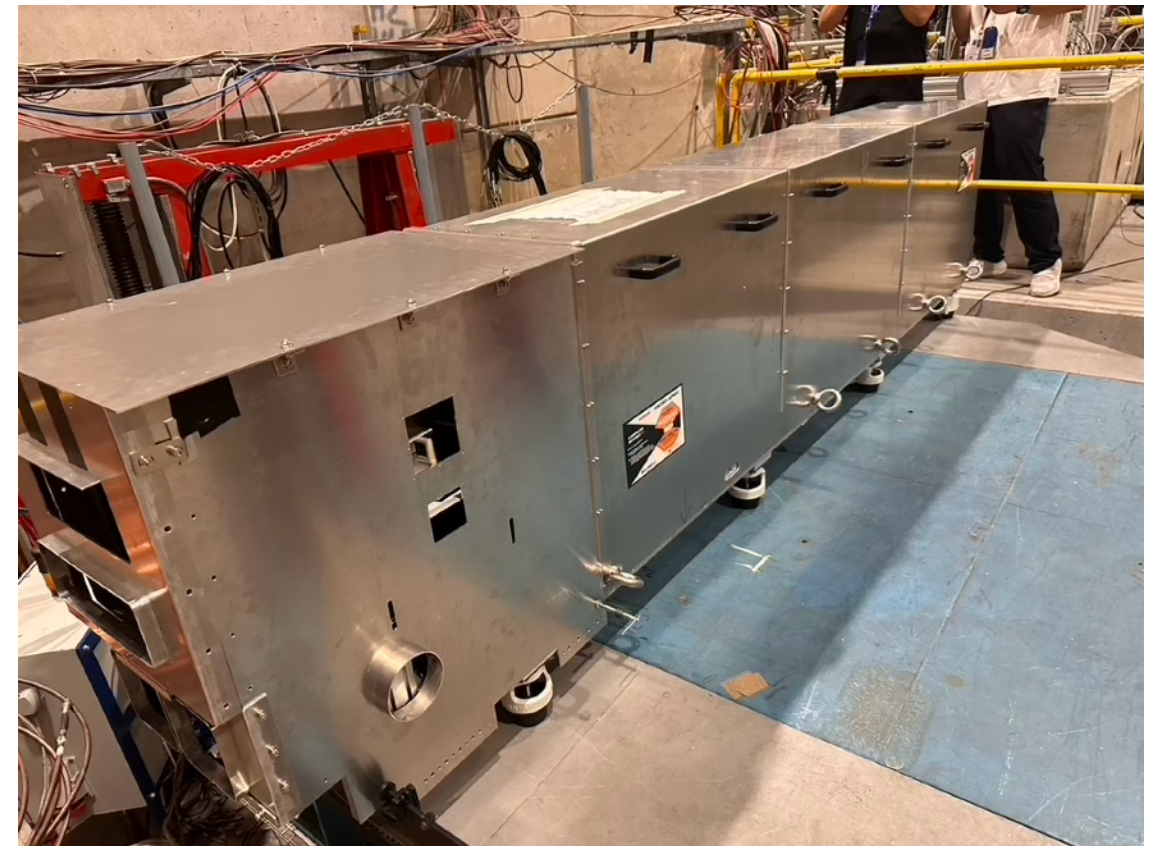


• DAQ system:

Same as 2022 test beam experiment

Installation : TB2024

- We completed the full-size module and installed it in the experimental hall
 - Calibration is currently in progress and data is being collected!!!

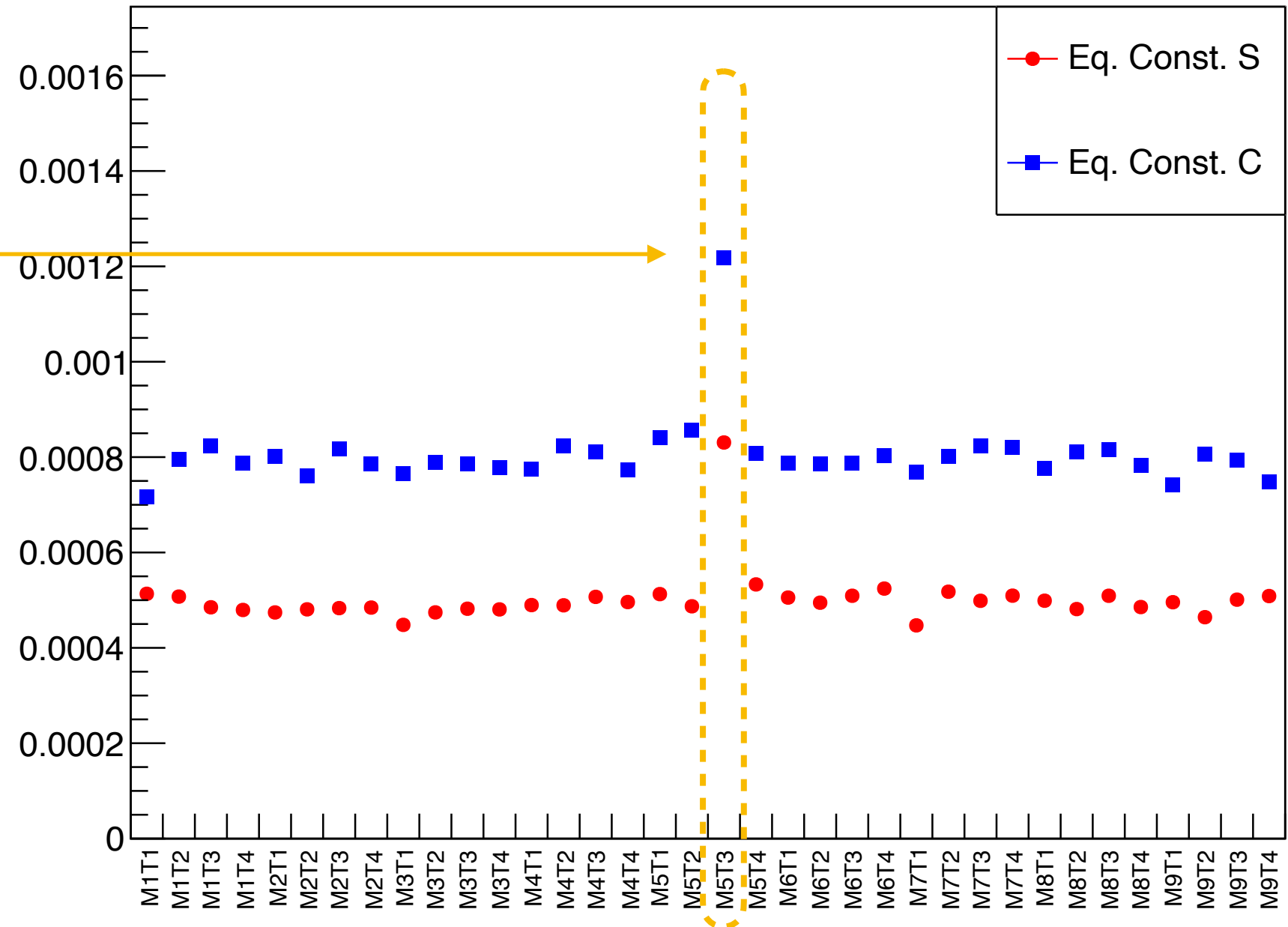


Equalization Constant

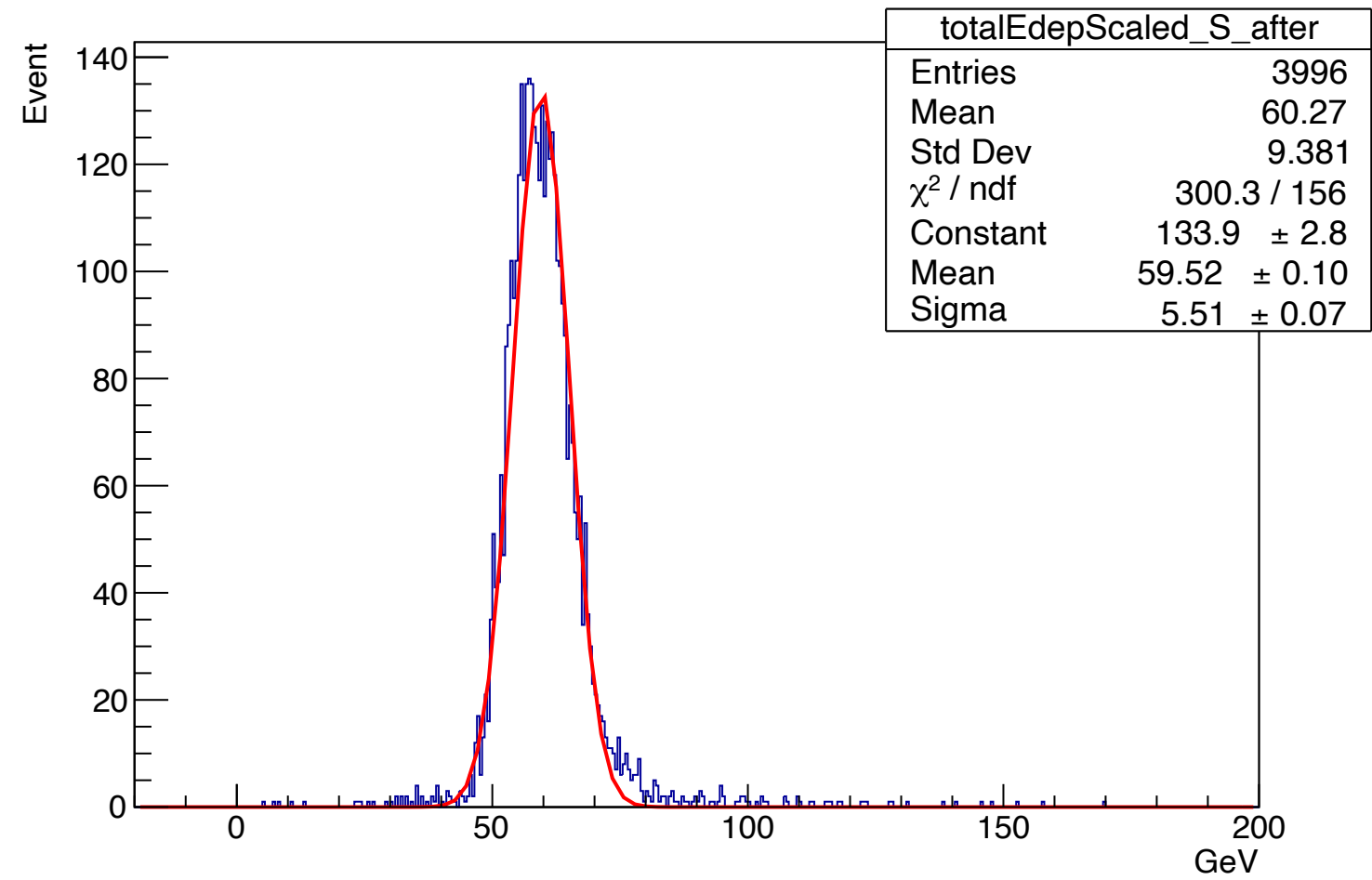
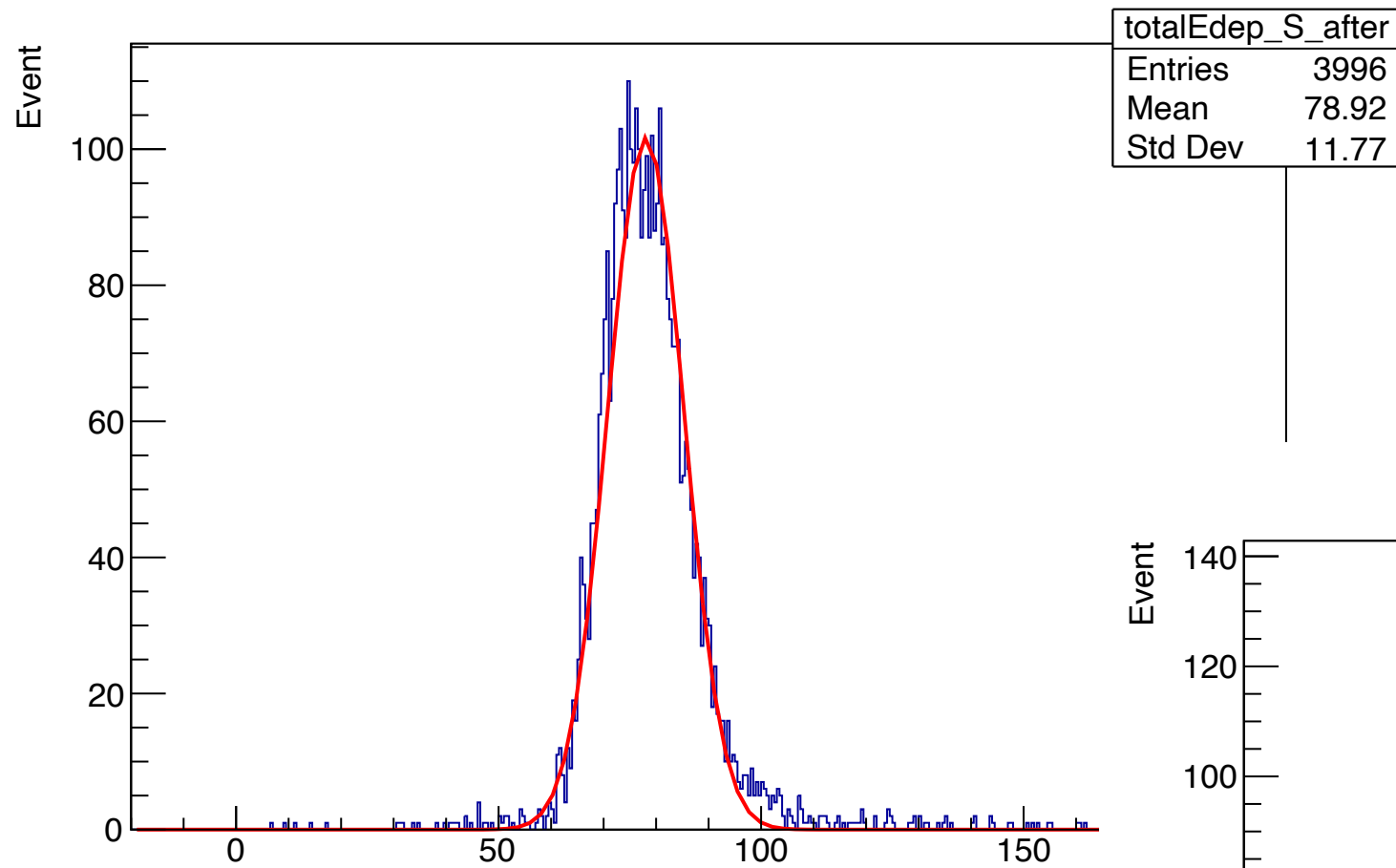
Very preliminary results !!!!

Overlay of Two Histograms

MCP PMT Tower

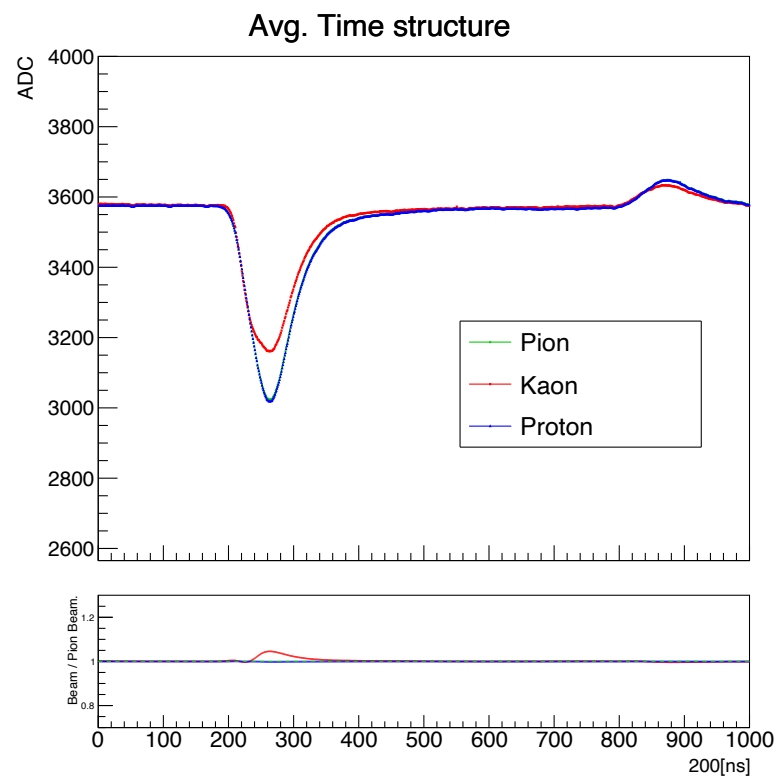


Calibrated Energy as 60 GeV (e^+)

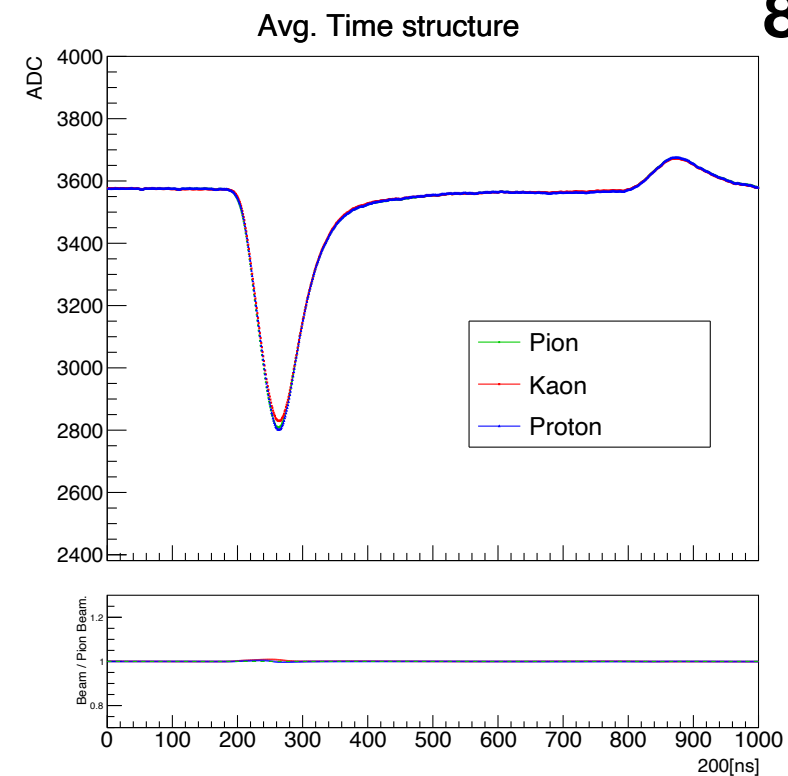


Avg Time Structure (Scintillator)

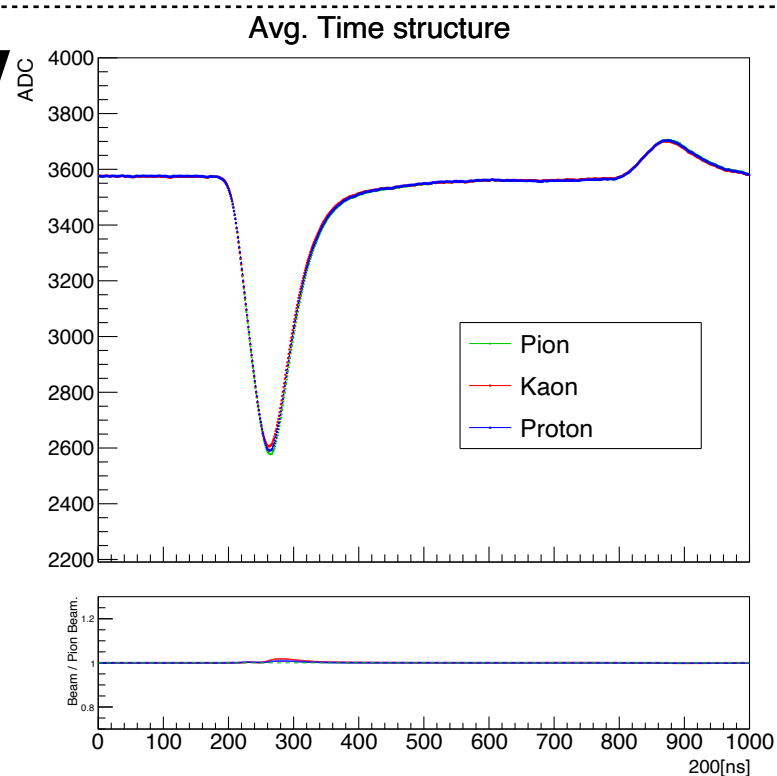
60 GeV



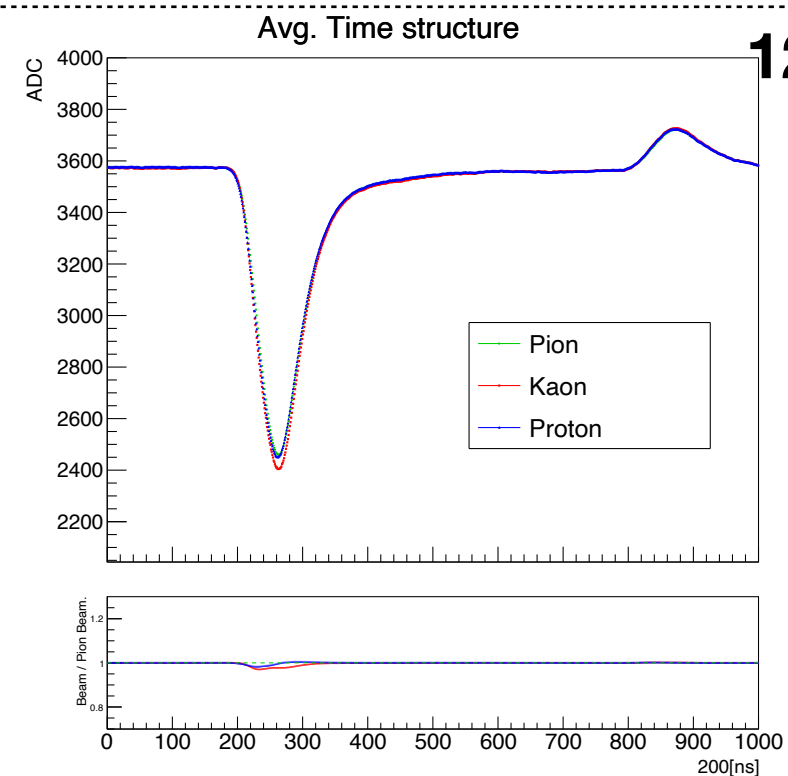
80 GeV



100 GeV



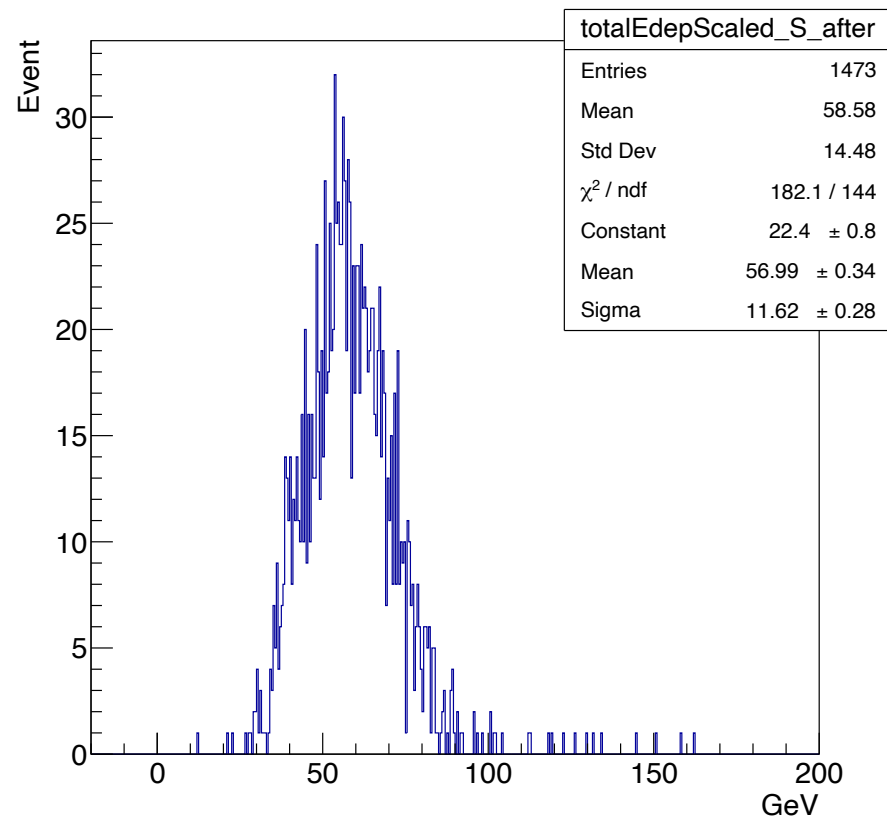
120 GeV



Energy Distribution (S, 80GeV)

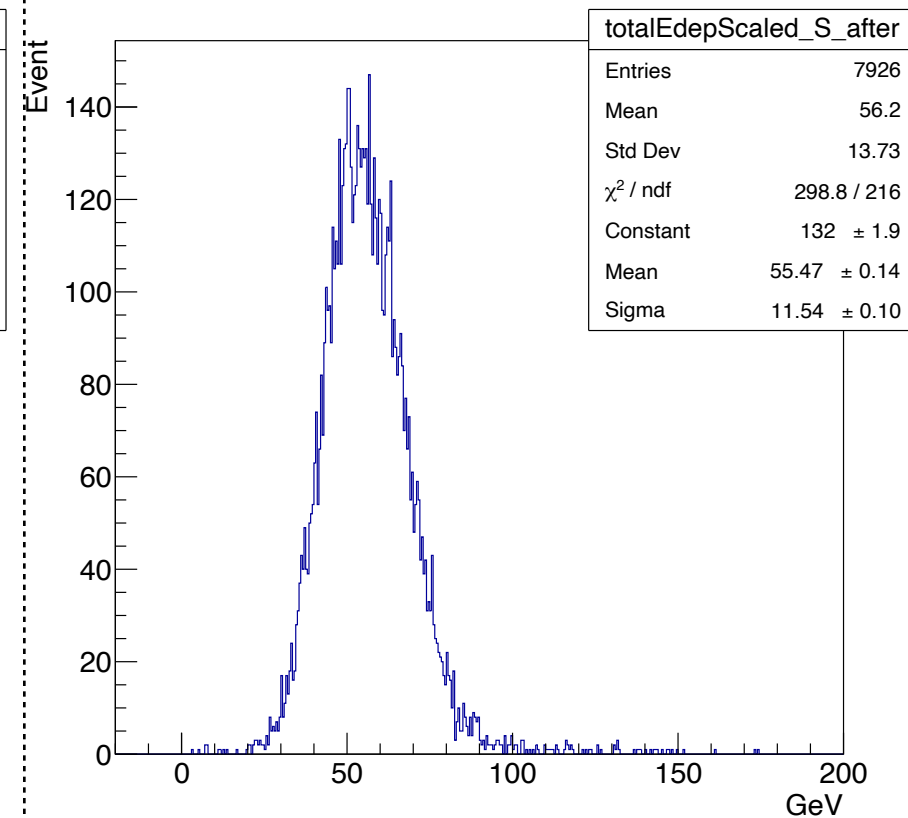
Very preliminary results !!!!

Kaon



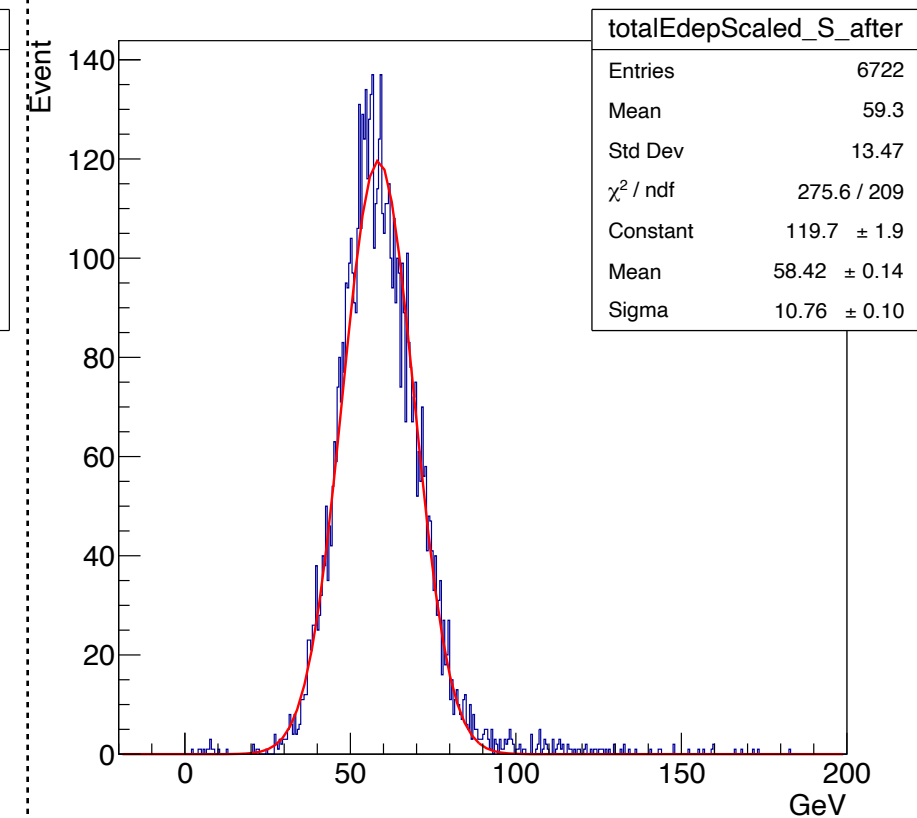
rms/mean = 24.7 %

Proton



rms/mean = 24.4 %

Pion

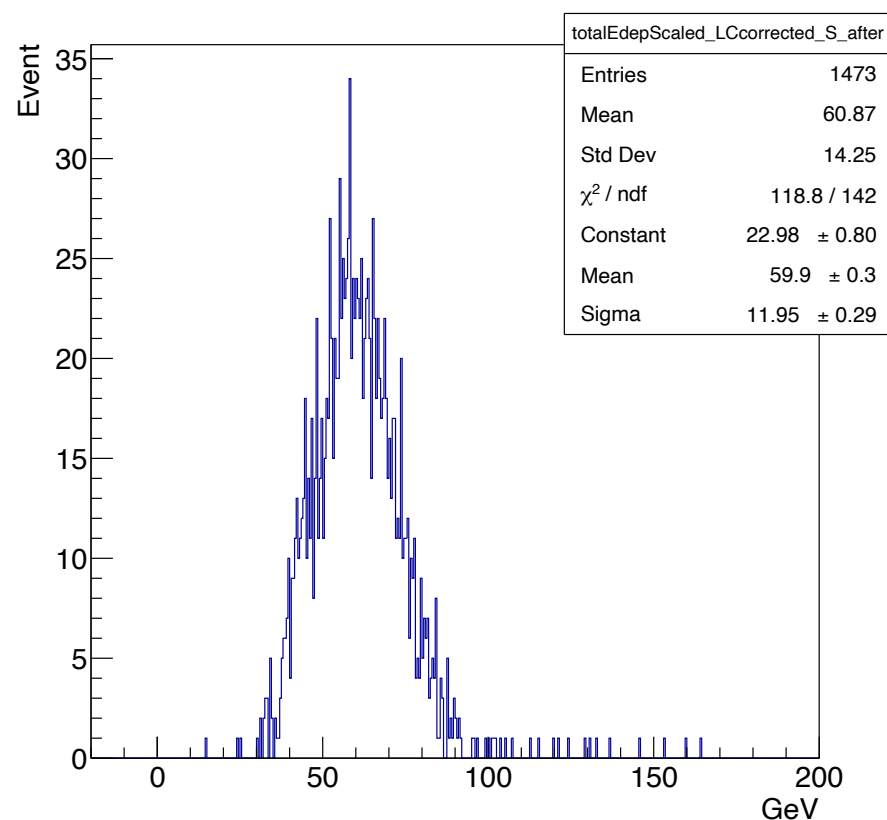


rms/mean = 22.7 %

Energy Distribution (S, LC correction, 80GeV)

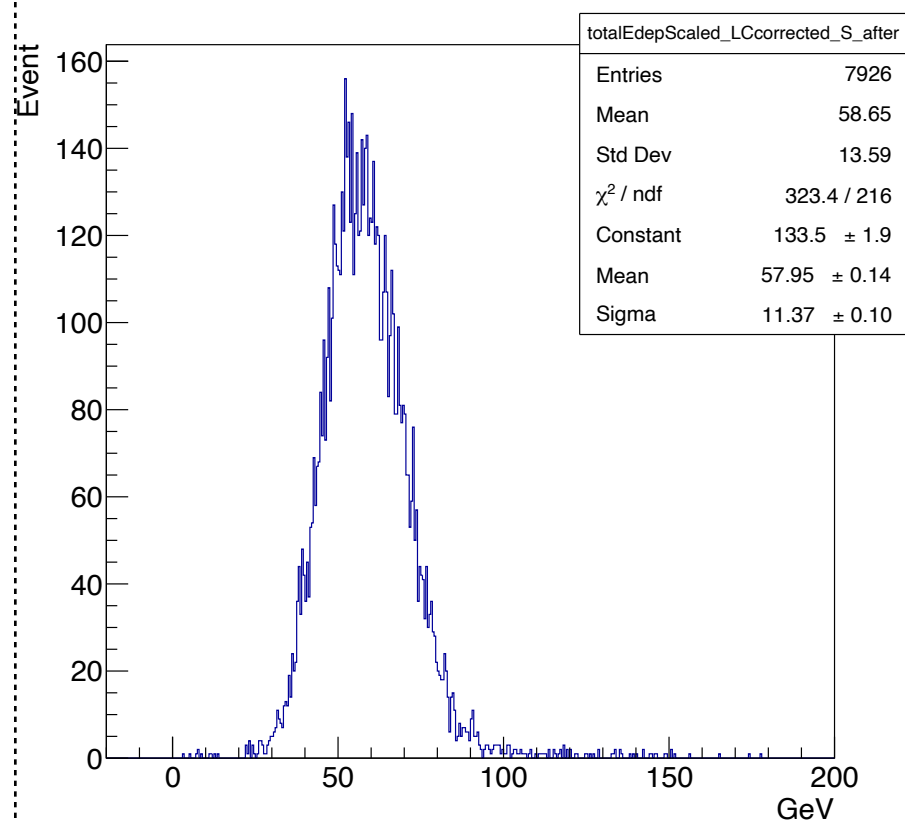
Very preliminary results !!!!

Kaon



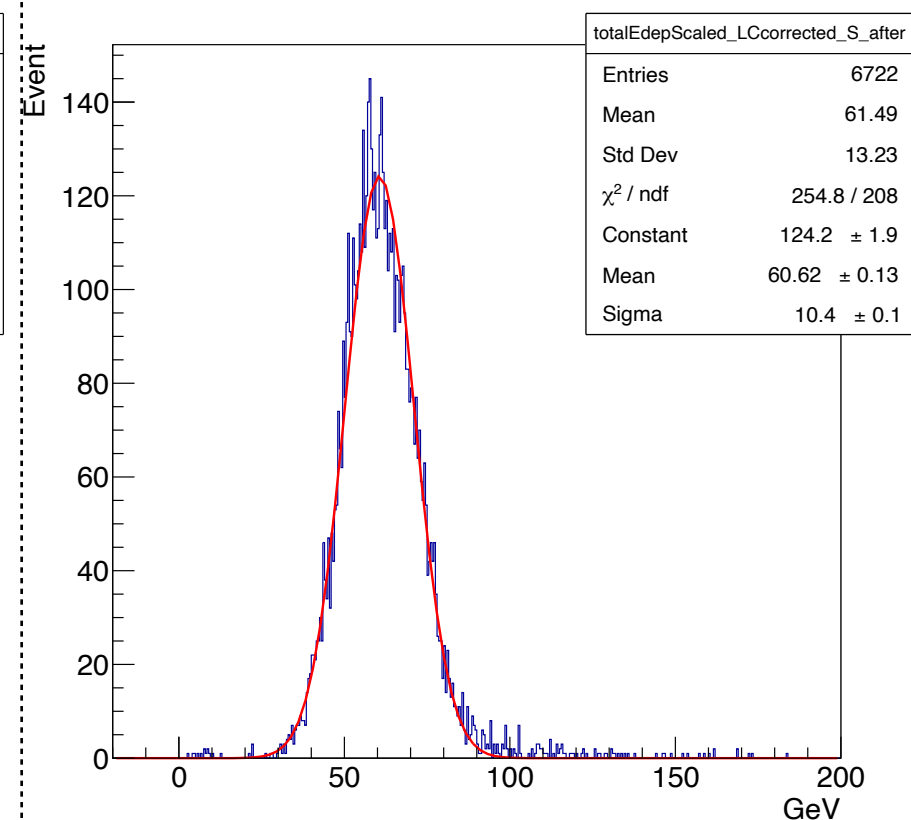
rms/mean = 23.4 %

Proton



rms/mean = 23.2 %

Pion



rms/mean = 21.5 %

Leakage Counter

- Why:
 - Leakage counters mitigate the effects of **shower leakage** in hadron calorimeter measurements.
 - They improve **energy resolution** by accounting for energy that escapes the calorimeter.
- How It Works:
 - **Scintillating plastic counters** detect energy exiting the calorimeter
 - These counters measure the portion of the **hadronic shower** that is not fully contained
 - Signals from leakage counters are used to **correct the scintillation signal**
- Important facts:
 - Hadronic showers, especially at high energies, may not be fully contained due to the large volume required
 - By incorporating leakage energy into the total signal, **energy fluctuations** caused by incomplete containment are reduced
 - This leads to **more accurate overall energy measurements**

Dual-readout Correction

- Concept:
 - Hadronic showers contain both **electromagnetic (em)** and **non-electromagnetic (non-em)** components.
 - Non-relativistic particles (e.g., protons) don't produce **Cherenkov light**, allowing separation of em and non-em showers.
- Dual Readout:
 - **Scintillation signal (S)**: Measures visible energy (total energy including **em and non-em components**)
 - **Cherenkov signal (C)**: Measures primarily **electromagnetic energy**.
- Energy Calculation:
 - By comparing the two signals, the **electromagnetic fraction** f_{em} can be determined.
 - Formula to calculate total energy:
$$E = \frac{S - \chi C}{1 - \chi}$$
: Constant depending on calorimeter characteristics.
- Advantage:
 - Reduces fluctuations in f_{em} , leading to more precise energy measurement